

Finite Golden Seeds and Uniform Analytic Transitions

Multiplicative ladders, real-order signed kernels,
reflected log-gamma moments, and a sharp depth obstruction

A research continuation for the ProveIt project
Prepared with OpenAI assistance

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Abstract

We continue four lines of research in the current ProveIt polylogarithm manuscript, distinguishing analytic identities, exact formal relations, and numerical evidence. First, primitive-divisor theory reduces every finite multiplicative seed for a positive real quadratic unit to indices at most twelve or twenty-four, according to its norm. At the golden reciprocal we determine the complete integral lattice: six generators for all positive powers, four for even powers, and no rational seeds after restricting the base to its fifth or any higher integer power. This proves completeness of a historically known even-power alphabet, without asserting sufficiency for higher-weight period identities.

Second, the coefficients $H_{n-1}^{(b)}/n^a$, with real $a, b > 0$, are moments of a finite signed measure exactly when $a + b \geq 1$. Above the boundary the density has one simple sign change; on the boundary it is strictly negative and a positive atom appears at one. The measure gives a unique simple angular zero throughout the upper half unit disk and a strictly increasing Cartesian abscissa. We also determine the near-critical density crossing and the concentration law of its positive mass.

Third, reflected log-gamma moments have a uniform transition at $n/(m+1) = \log m + O(1)$, with normalized limit $\exp(ce^{-s})$, where $c = \zeta(2)/(2\gamma) - \gamma$. We prove the first correction, an expansion to every fixed order, and exact identities for near-diagonal residue polynomials. Finally, a determinant-one certificate proves that the weight-six binary shuffle quotient has exactly one direction beyond all depth-two words and their six standard argument transforms. Two explicit quotient identities follow. The existing S_6 special-value conjecture remains open. The accompanying package supplies exact arithmetic replays, independent analytic audits, numerical diagnostics, and proposed editorial corrections.

Source snapshot. [ProveIt manuscript, commit 28357e8ca63dd78327db91d9be239d75e4462879](#).
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1 Scope, results, and mathematical status

The pinned source is already a substantial research manuscript [1]. It proves the S_4 reduction, several all-weight Gaussian and complementary-depth statements, algebraic low-weight ladders, and a detailed fixed-reflected-exponent asymptotic theory. We address four gaps: unbounded support of multiplicative seeds, real orders and the boundary of signed-measure representability, simultaneous growth of reflected exponents, and the exact limitation of a natural depth-reduction vocabulary.

The source has several different notions of a relation. A relation among algebraic units can be checked in a number field. A relation in a formal shuffle or exterior-symbol quotient can be certified by exact algebra. A numerical identity among special values requires an analytic derivation or another valid equality proof. These notions interact, but the implication from a formal relation to a numerical period identity is not automatic. Every result below states the space in which it is proved.

1.1 A guide to the contributions

Question	Result in this article	Nature of the conclusion
Can larger golden exponents supply new multiplicative seeds?	All finite relations have six explicit integral generators; the even-power lattice has four.	Unbounded arithmetic completeness, Theorem 2.2.
Which real orders admit a finite signed moment measure?	Exactly $a + b \geq 1$; the critical line has an endpoint atom.	Analytic classification, Theorem 3.1.
Does the semicircle zero extend inside the disk?	Exactly one simple angular zero at every radius; its abscissa is strictly increasing.	Analytic geometry, Theorem 3.3.
How does the critical atom form?	Explicit crossing asymptotics and an exponential law on a logarithmically rescaled coordinate.	Uniform boundary analysis, Section 4.
When does the fixed- m moment equivalent cease to be uniform?	A nontrivial exponential profile occurs at $n/(m+1) - \log m = O(1)$.	Uniform asymptotic theorem, Theorem 5.1.
Can six standard argument changes remove all weight-six triples?	Their depth-two span has codimension exactly one; two triple combinations have explicit reductions.	Formal quotient theorem, Theorem 6.1.

The golden alphabet itself has historical precedent in Bailey–Broadhurst [3]. The complete arithmetic argument here uses Flatters’ later primitive-divisor theorem [2]. The log-gamma family is classical and is discussed by Amdeberhan et al. [5]; the new conclusion concerns a growing second exponent in the regime explicitly left open by the source. Li’s recent positive-measure treatment of star polylogarithms [6] provides nearby geometric context, while our measure is signed and our indices are real. Claims of extension in this article are relative to the pinned manuscript and these identified sources. A complete literature search for every consequence, or a claim of universal priority, is not intended.

1.2 Conventions and proof architecture

For $|z| < 1$ we use the strict convention

$$\mathrm{Li}_{s_1, \dots, s_d}(z, 1, \dots, 1) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{s_1} \dots n_d^{s_d}}.$$

When exponents are real, the positive integer powers in the denominators use the real logarithm. The notation $\zeta(s, x)$ denotes the Hurwitz zeta function and γ Euler's constant; standard gamma and zeta identities use the conventions of the NIST DLMF [8]. For boundary values of the real-order depth-two function, analytic continuation on $\mathbb{C} \setminus [1, \infty)$, equivalently radial Abel limits away from 1, is used when the original series fails to converge. Logarithms in the one-variable hyperlogarithm identities are continued consistently from $0 < z < 1$.

The arithmetic proof first rules out all sufficiently large indices, then solves a finite integral lattice with explicit prime-ideal witnesses. This order is essential: a finite search alone would not prove completeness. The signed-kernel proof preserves the cancellation at the origin of its defining integral. At the critical line the usual Fubini argument is replaced by a differentiated Laplace identity, which determines the missing atom. The moment proof similarly avoids a tempting unjustified interchange: its uniform limit comes from a weighted gamma-distribution integral, not a summation of the divergent fixed- m pole expansion. The depth theorem specifies its formal quotient before using exact ranks.

The analytic proofs have been independently reviewed within this research workflow. Their review notes and finite arithmetic receipts are included in the package. This is not formal verification in a proof assistant or a substitute for external mathematical review. Floating-point checks are diagnostics throughout; none is used to promote a conjectural special-value identity to a theorem.

2 Complete multiplicative seeds at the golden base

The complementary-power classification in the current manuscript [1] concerns relations of the form $r^a + r^b = 1$. Golden ladders also use products of many factors $1 - r^n$. We now give a completeness theorem for all such finite multiplicative relations at the golden base, without imposing an exponent bound in advance. The result determines an integral relation lattice, explains the eight arguments in the manuscript's higher golden formulas, and classifies which integer powers of the base retain any such seeds.

The four exceptional indices 1, 6, 10, 12, when the base is the square of the golden reciprocal, were already recorded by Bailey and Broadhurst [3, §2, preceding (22)]. We do not claim discovery of that list. The contribution here is its unbounded completeness proof, an explicit integral lattice for all positive powers, and its consequences for power restrictions and exterior symbols. The arithmetic input is Flatters' later theorem on primitive divisors [2].

2.1 An effective bound for every real quadratic unit

Let K be a real quadratic field, let σ denote its nontrivial automorphism, and let $r \in \mathcal{O}_K^\times$ satisfy $0 < r < 1$. Put $V_n(r) = 1 - r^n$ for $n \geq 1$. A prime ideal $\mathfrak{p}$ is *primitive* for $V_n(r)$ if it divides $V_n(r)$ and none of $V_1(r), \dots, V_{n-1}(r)$. Because r is a unit, this says exactly that its residue in the finite field $\mathcal{O}_K/\mathfrak{p}$ has multiplicative order n .

We use the following part of Flatters' theorem: if u is a positive real quadratic unit of norm $+1$, then $N_{K/\mathbb{Q}}(u^m - 1)$ has a primitive rational prime divisor for every $m > 12$ [2, Theorem 1.4]. Here a rational prime is primitive if it divides this norm and no earlier nonzero norm. The following passage to prime ideals is needed when the original unit has norm -1 .

Theorem 2.1 (Uniform quadratic cutoff). *If $N_{K/\mathbb{Q}}(r) = +1$, then $1 - r^n$ has a primitive prime-ideal divisor for every $n > 12$. If $N_{K/\mathbb{Q}}(r) = -1$, it has one for every $n > 24$.*

Consequently, in any finite relation

$$\prod_{n \geq 1} (1 - r^n)^{c_n} = r^b, \quad c_n, b \in \mathbb{Z}, \quad (2.1)$$

all coefficients above 12 vanish in the norm +1 case, and all coefficients above 24 vanish in the norm -1 case. The same support conclusion holds for rational exponents.

Proof. For norm +1, apply Flatters' theorem to $u = r^{-1} > 1$. At index $n > 12$, choose a primitive rational divisor p of $N_{K/\mathbb{Q}}(u^n - 1)$ and a prime ideal $\mathfrak{p}$ above p dividing $u^n - 1$. The residue order of u is exactly n : an order $j < n$ would make p divide the earlier norm $N_{K/\mathbb{Q}}(u^j - 1)$. Inversion preserves order, so r also has order n at this ideal.

For norm -1, put $q = r^2$ and apply the norm +1 theorem to $u = r^{-2}$. For each $m > 12$, we obtain a primitive rational divisor of $N_{K/\mathbb{Q}}(q^m - 1)$, because $u^j - 1 = -q^{-j}(q^j - 1)$ differs by a unit. At a suitable ideal $\mathfrak{p}$ above this prime, the residue of q has exact order m .

The residue characteristic is odd. Indeed, the residue field of an ideal above 2 in a quadratic field has size 2 or 4; the residue of any unit therefore has order 1 or 3. Thus 2 divides some norm $N_{K/\mathbb{Q}}(q^j - 1)$ with $j \leq 3$, and cannot be primitive at $m > 12$.

Let a be the residue of r at $\mathfrak{p}$. Since a^2 has order m , the order of a is either m or $2m$, and it must be $2m$ when m is even. The conjugation formula

$$\sigma(r) = -r^{-1} \quad (2.2)$$

identifies the residue of r at the conjugate ideal, under the induced field isomorphism, with $-a^{-1}$. When m is odd, these two residues have orders m and $2m$ in some order. To check this directly, if a has odd order m , then $-a^{-1}$ has order $2m$. If a has order $2m$, then $a^m = -1$ in odd characteristic, so $(-a^{-1})^m = 1$; its square still has order m , forcing its order to be m . In this odd- m case the ideals must be distinct: if they coincided, the induced residue-field automorphism would preserve order. Thus no splitting assumption is needed.

For an even target $n > 24$, take $m = n/2 > 12$. If m is even, the chosen ideal gives residue order n . If m is odd, the chosen ideal or its conjugate gives order $2m = n$. For an odd target $n > 24$, take $m = n$ and choose the ideal giving order $m = n$. This proves the prime-ideal statement in every case.

Finally, suppose (2.1) has a nonzero coefficient above the relevant cutoff, and choose its largest supported index N . At a primitive ideal for $V_N(r)$, all lower factors and the unit r have valuation zero. Taking valuations gives $c_N v_{\mathfrak{p}}(V_N(r)) = 0$, which contradicts both $c_N \neq 0$ and $v_{\mathfrak{p}}(V_N(r)) > 0$. Clearing denominators proves the rational-exponent version. $\square$

This theorem bounds multiplicative seeds. It does not place an unconditional bound on the weight of all polylogarithmic identities. For a fixed quadratic base, it reduces seed determination to finitely many exact ideal valuations and a unit calculation.

2.2 The full integral lattice at the golden reciprocal

Set

$$\rho = \frac{\sqrt{5} - 1}{2}, \quad V_n = 1 - \rho^n, \quad K = \mathbb{Q}(\sqrt{5}), \quad \mathcal{O}_K = \mathbb{Z}[\rho].$$

All V_n are positive, so the following equations have no real logarithm or sign ambiguity.

Theorem 2.2 (Complete golden seed lattice). *The group of finite integer relations*

$$\prod_{n \geq 1} V_n^{c_n} = \rho^b, \quad (c_n) \in \mathbb{Z}^{\langle \mathbb{Z}_{>0} \rangle}, \quad b \in \mathbb{Z}, \quad (2.3)$$

is free of rank six. An integral basis is

$$\begin{aligned} V_1 &= \rho^2, & V_2 &= \rho, \\ V_6 &= \rho^{-1}V_3^2, & V_{12} &= \rho^{-2}V_4V_3^3, \\ V_{20} &= \rho^{-1}V_4^3V_{10}, & V_{24}V_4 &= \rho^{-2}V_3^4V_8^2. \end{aligned} \quad (2.4)$$

Equivalently, c occurs in (2.3) precisely when its support is contained in

$$S = \{1, 2, 3, 4, 6, 8, 10, 12, 20, 24\} \quad (2.5)$$

and

$$\begin{aligned} c_3 + 2c_6 + 3c_{12} + 4c_{24} &= 0, \\ c_4 + c_{12} + 3c_{20} - c_{24} &= 0, \\ c_8 + 2c_{24} &= 0, \\ c_{10} + c_{20} &= 0. \end{aligned} \quad (2.6)$$

The uniquely determined unit exponent is

$$b = 2c_1 + c_2 - c_6 - 2c_{12} - c_{20} - 2c_{24}. \quad (2.7)$$

We first record the exact finite information needed in the proof. Reducing powers using $\rho^2 = 1 - \rho$ gives

n	V_n	n	V_n
1	ρ^2	2	ρ
3	$2\rho^2$	4	$\sqrt{5}\rho^2$
6	$4\rho^3$	8	$3\sqrt{5}\rho^4$
10	$11\rho^5$	12	$8\sqrt{5}\rho^6$
20	$55\sqrt{5}\rho^{10}$	24	$144\sqrt{5}\rho^{12}$

Substitution proves every equation in (2.4). In particular, the indices

$$E = \{1, 2, 6, 12, 20, 24\}$$

introduce no prime ideal absent from lower indices.

Every other index through 24 has the following primitive witness. A scalar a denotes the ideal $(p, \rho - a)$ with residue field $\mathbb{F}_p$. The entry θ denotes the ideal (p) with residue field $\mathbb{F}_p[\theta]/(\theta^2 + \theta - 1)$; the defining polynomial is irreducible for the three primes $p = 2, 3, 7$ used below. All powers in the last column are computed in the indicated residue field.

n	p	residue of ρ	proper-power checks
3	2	θ	$\theta^1 = \theta$
4	5	2	$2^2 = 4$
5	11	3	$3^1 = 3$
7	29	23	$23^1 = 23$
8	3	θ	$\theta^4 = 2$
9	19	4	$4^3 = 7$
10	11	7	$7^5 = 10, \quad 7^2 = 5$
11	199	61	$61^1 = 61$
13	521	421	$421^1 = 421$
14	29	5	$5^7 = 28, \quad 5^2 = 25$
15	31	18	$18^5 = 25, \quad 18^3 = 4$
16	7	θ	$\theta^8 = 6$
17	3571	1103	$1103^1 = 1103$
18	19	14	$14^9 = 18, \quad 14^6 = 7$
19	9349	7563	$7563^1 = 7563$
21	211	178	$178^7 = 196, \quad 178^3 = 144$
22	199	137	$137^{11} = 198, \quad 137^2 = 63$
23	139	63	$63^1 = 63$

For each scalar row, direct arithmetic verifies $a^2 + a - 1 = 0$ and $a^n = 1$ modulo p ; for each quadratic row it verifies $\theta^n = 1$. The last column lists the values at n/ℓ for every prime $\ell \mid n$, and none equals 1. Thus the order is exactly n . The listed rational primes are verified by trial division, and irreducibility of the three quadratic polynomials is checked by testing their residues. The accompanying script replays all of these finite checks in exact arithmetic.

Proof of Theorem 2.2. Theorem 2.1 shows that every relation (2.3) is supported on indices at most 24. Subtract integer multiples of the six rows in (2.4) to set the coefficients at 1, 2, 6, 12, 20, 24 to zero. Each row has coefficient +1 in its own distinct pivot column and zero in every other pivot column, so this elimination is integral and leaves all eliminated coefficients zero.

If the residual relation is nonzero, let N be its largest supported index. Then $N \notin E$ and the table supplies an ideal primitive for V_N . Every lower factor and ρ has valuation zero there, whereas V_N has positive valuation. Applying this valuation again contradicts the nonzero coefficient at N . Hence the six rows span the entire relation group over $\mathbb{Z}$.

Their distinct pivot columns also prove independence. Reading their other columns gives (2.6), and reading their unit exponents gives (2.7). The unit exponent is unique because ρ is not a root of unity. $\square$

Remark 2.3 (Why the ideal witnesses matter). The norms of V_5 and V_{10} are 11 and -121 . One ideal above 11 has residue $\rho = 3$ of order 5; its conjugate has residue $\rho = 7$ of order 10. Thus V_{10} has a primitive prime-ideal divisor even though its norm has no primitive rational prime divisor. A proof using only new rational prime factors of the norms would miss this constraint.

2.3 Four even-power seeds and an integral parity condition

Put $q = \rho^2$ and $U_j = 1 - q^j = V_{2j}$.

Corollary 2.4 (The complete even-power lattice). *All finite integral relations $\prod_{j \geq 1} U_j^{c_j} = \rho^b$ are*

generated freely by

$$\begin{aligned} U_1 &= \rho, \\ U_6^2 &= \rho^{-1} U_2^2 U_3^3, \\ U_{10} &= \rho^{-1} U_2^3 U_5, \\ U_{12} U_2 &= U_3^2 U_4^2. \end{aligned} \tag{2.8}$$

Their support is contained in

$$\{1, 2, 3, 4, 5, 6, 10, 12\} \quad \text{in the } q \text{ index.} \tag{2.9}$$

If the four rows are used with multiplicities (A, B, C, D) , the resulting exponent of ρ is $A - B - C$. The result is an integer power of q precisely when $A - B - C$ is even.

Proof. Write the multiplicities of the rows in (2.4) as $a_1, a_2, a_6, a_{12}, a_{20}, a_{24}$. Their odd-index coefficients vanish exactly when

$$a_1 = 0, \quad 2a_6 + 3a_{12} + 4a_{24} = 0.$$

Thus a_{12} is even and $a_6 = -3(a_{12}/2) - 2a_{24}$. The four free integer parameters $a_2, a_{12}/2, a_{20}, a_{24}$ give exactly (2.8). The unit exponent follows from the four displayed identities, proving the parity assertion. $\square$

The support (2.9) is precisely the eight-argument set

$$\mathcal{A} = \{\rho^2, \rho^4, \rho^6, \rho^8, \rho^{10}, \rho^{12}, \rho^{20}, \rho^{24}\} \tag{2.10}$$

used by the manuscript's conjectured higher-weight golden ladders. The theorem now establishes that no arbitrarily large even exponent can introduce another multiplicative seed. At the level of rational logarithmic relations, the integral parity condition disappears because $\mathbb{Q} \log q = \mathbb{Q} \log \rho$.

2.4 Which powers of the base retain any seeds?

For a positive integer h , define the rational relation space

$$\mathcal{R}_h = \left\{ (c_n) \in \mathbb{Q}^{(\mathbb{Z}_{>0})} : \sum_{n \geq 1} c_n \log(1 - \rho^{hn}) \in \mathbb{Q} \log \rho \right\}. \tag{2.11}$$

One may replace the line $\mathbb{Q} \log \rho$ by $\mathbb{Q} \log(\rho^h)$ without changing this space.

Theorem 2.5 (Sharp power-restriction ranks). *For every positive integer h ,*

$$\dim_{\mathbb{Q}} \mathcal{R}_h = \begin{cases} 6, & h = 1, \\ 4, & h = 2, \\ 1, & h = 3, 4, \\ 0, & h \geq 5. \end{cases} \tag{2.12}$$

For $h = 3$, the unique rational direction is $V_6/V_3^2 = \rho^{-1}$. For $h = 4$, it is

$$\frac{V_{12}^4 V_8^6}{V_4^7 V_{24}^3} = \rho^{-2}. \tag{2.13}$$

The primitive integral relations whose right sides are integer powers of their own base $r = \rho^h$ are

$$h = 3: \quad (1 - r^2)^3 = r^{-1}(1 - r)^6, \tag{2.14}$$

$$h = 4: \quad (1 - r^3)^8(1 - r^2)^{12} = r^{-1}(1 - r)^{14}(1 - r^6)^6. \tag{2.15}$$

Proof. Clear denominators in (2.11) and exponentiate. The resulting relation has the form (2.3) and is supported only on multiples of h . Theorem 2.2 supplies the complete support and constraints.

The cases $h = 1, 2$ follow immediately from that theorem and Corollary 2.4. For $h = 3$, the permitted support is $\{3, 6, 12, 24\}$. The equation at index 8 in (2.6) forces $c_{24} = 0$; the equation at index 4 then forces $c_{12} = 0$. The remaining freedom is $c_3 = -2c_6$.

For $h = 4$, the permitted support is $\{4, 8, 12, 20, 24\}$. The equation at index 10 forces $c_{20} = 0$; that at index 3 gives $3c_{12} + 4c_{24} = 0$. Thus the integer solutions are

$$c_{12} = 4t, \quad c_{24} = -3t, \quad c_4 = -7t, \quad c_8 = 6t,$$

which give (2.13). Their unit exponent is $-2t$, so it is an integer power of ρ^4 precisely when t is even. For $h = 3$, the unit exponent is $-c_6$, so an integer power of ρ^3 requires $3 \mid c_6$. These conditions give the two primitive integral relations displayed in the theorem.

For $h \geq 5$, support is empty unless h divides a member of S in (2.5). The possibilities are $h \in \{5, 6, 8, 10, 12, 20, 24\}$. It suffices to exclude 5, 6, 8, as their multiples impose further restrictions. For $h = 5$, only indices 10, 20 are permitted; the index-4 equation forces $c_{20} = 0$, and the index-10 equation then forces $c_{10} = 0$. For $h = 6$, only 6, 12, 24 are permitted; the equations at indices 8, 4, 3 eliminate them in that order. For $h = 8$, only 8, 24 are permitted; the equation at index 4 eliminates 24, then that at index 8 eliminates 8. This proves all cases of (2.12). $\square$

The number field remains $\mathbb{Q}(\sqrt{5})$ under these power substitutions, but the permitted positive-power seed space can shrink from dimension six to zero. Thus the field alone does not determine the seed count for a specified base and argument family.

2.5 All-weight kernels of the exterior-symbol map

Put

$$W = K^\times \otimes_{\mathbb{Z}} \mathbb{Q}, \quad \langle x \rangle = x \otimes 1 \in W, \quad v = \langle \rho \rangle,$$

and use additive notation for W . For $w \geq 2$, define the rationalized classical exterior-symbol map

$$\beta_w([x]) = \langle x \rangle^{\otimes(w-2)} \otimes (\langle x \rangle \wedge \langle 1 - x \rangle) \quad (x \neq 0, 1), \quad (2.16)$$

with target $W^{\otimes(w-2)} \otimes \wedge^2 W$. Here $[x]$ denotes a generator in a free rational vector space. We make no identification of its quotient with numerical periods.

Theorem 2.6 (Complete positive-power exterior-symbol kernels). *On the free span of $[\rho^n]$, $n \geq 1$, the kernel of β_w has dimension six at every weight $w \geq 2$. An explicit basis consists of*

$$\sum_n \frac{c_n}{n^{w-1}} [\rho^n], \quad (2.17)$$

where c runs through the six coefficient vectors in (2.4). Every kernel element is supported on (2.5). On even powers the kernel has dimension four and is supported on (2.10). On the span of $[\rho^{hn}]$, its dimensions are (2.12).

Proof. The vector v is nonzero because ρ is not torsion. Multilinearity gives, for every finitely supported rational vector a ,

$$\beta_w\left(\sum_n a_n [\rho^n]\right) = v^{\otimes(w-2)} \otimes \left(v \wedge \sum_n a_n n^{w-1} \langle V_n \rangle\right). \quad (2.18)$$

Tensoring over a field with a fixed nonzero pure tensor is injective, and $v \wedge z = 0$ if and only if $z \in \mathbb{Q}v$. Consequently (2.18) vanishes exactly when

$$\sum_n a_n n^{w-1} \langle V_n \rangle \in \mathbb{Q}v. \quad (2.19)$$

After clearing denominators, this says that a positive real algebraic number $\prod_n V_n^{c_n} \rho^{-b}$ is torsion. A positive real root of unity is 1, so the condition is equivalent to the actual multiplicative relation (2.3), with the coefficients proportional to $a_n n^{w-1}$. The diagonal operation $a_n \mapsto n^{w-1} a_n$ is invertible and preserves support. Theorems 2.2 and 2.5 therefore give the basis, support, and every stated dimension. $\square$

Scope of the obstruction. This theorem computes the kernel of the explicitly specified algebraic map. A nonzero exterior symbol rules out a certificate requiring vanishing in this map. Vanishing in the number-field symbol alone does not supply a rational-function deformation, lower-weight compatibility, or the regulator information needed to prove a higher polylogarithm identity. In particular, dimension four does not assert that four numerical ladders exist at every weight, and it does not prove that none exists above weight nine. Numerical period independence remains a separate arithmetic question.

The constructive consequence is that searches beginning with this symbol can use (2.10) as a complete even-power alphabet and work in four weighted seed coordinates. Increasing the positive even exponents beyond 24 cannot enlarge that particular kernel.

2.6 Exact replay and next proof problems

The file `code/golden_seed_certificate.py` uses only Python's standard library. It verifies the six generator identities in $\mathbb{Q}[X]/(X^2 + X - 1)$, the two primitive own-base relations at $h = 3, 4$, all eighteen primitive prime-ideal witnesses, and the rational restriction ranks for $1 \leq h \leq 25$. The output is preserved in `data/golden_seed_certificate.json`. Every calculation in this replay uses exact integer or rational arithmetic. The infinite bounds rest on Flatters' theorem and the prime-ideal argument above; the proof for all $h \geq 5$ rests on the explicit support constraints.

Several further questions are now more sharply delimited. First, the unresolved golden formulas at weights five through nine can be approached in the complete eight-argument symbol alphabet; the missing step is an exact functional-equation or regulator certificate. Second, the same uniform 12/24 cutoff permits complete seed computations for other real quadratic units, retaining the unit index congruence when the base is not fundamental. Third, an effective prime-ideal cutoff for the cubic plastic and supergolden fields would give an analogous completeness theorem there. Finally, formalization can begin with the finite arithmetic interface: quadratic algebra, exact residue orders, integral elimination, and the tensor/exterior linear-algebra argument. The cited primitive-divisor theorem is an explicit separate dependency.

3 Real-weight signed kernels and a sharp measure threshold

The signed-kernel chapter of the pinned manuscript proves that the strict depth-two coefficients $H_{n-1}^{(b)}/n^a$, with integer $a \geq 1$ and real $b > 0$, have an integrable density with one sign change. That structure gives a unique zero of the imaginary part on the upper semicircle. The affine signed-moment continuation already permits real $a \geq 1$ for measure existence and total-variation estimates, but does not extend the one-crossing theorem to that range or treat $0 < a < 1$ [1]. We

now classify the complete domain of finite signed measures for these coefficients. Its boundary has a positive endpoint atom, and requires a different cancellation argument from the interior.

Hausdorff-Stieltjes methods for polylogarithmic geometry have substantial existing context. In particular, Li's recent work treats positive Hausdorff measures and likelihood-ratio ordering for integer-index multiple *star* polylogarithms [6]. Here the strict depth-two sequence has total moment mass zero, so its representing measure is signed. The results below concern real exponents, the sharp boundary of finite-measure representability, and the associated zero arc. The claim of extension is relative to the pinned manuscript; no complete priority claim is intended.

For $a, b > 0$ define

$$m_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad H_{n-1}^{(b)} = \sum_{j=1}^{n-1} j^{-b}, \quad F_{a,b}(z) = \sum_{n \geq 1} m_n(a, b) z^n \quad (|z| < 1). \quad (3.1)$$

The empty harmonic sum is zero. At integer exponents $F_{a,b}$ is the strict depth-two polylogarithm used in the source manuscript.

3.1 A real-order kernel with cancellation retained

For $T > 0$ put

$$k_{a,b}(T) = -\frac{T^{a+b-1}}{\Gamma(a+b)} + \frac{1}{\Gamma(a)\Gamma(b)} \int_0^\infty \frac{v^{b-1}}{e^v - 1} [T^{a-1} - (T-v)_+^{a-1}] dv, \quad (3.2)$$

and $\kappa_{a,b}(u) = k_{a,b}(-\log u)$ for $0 < u < 1$. Here $(T-v)_+^{a-1}$ means $(T-v)^{a-1}$ when $v < T$, and zero when $v > T$; its value at $v = T$ is irrelevant. In particular, if $0 < a < 1$, the left-hand singularity at $v = T$ is retained. The integral converges absolutely for each fixed T : its bracket is $O_T(v)$ at zero, the singularity at $v = T$ has integrable exponent $a-1 > -1$, and the tail decays exponentially. For $b \leq 1$ the two terms inside the bracket must remain combined near $v = 0$.

Theorem 3.1 (Sharp finite signed-measure classification). *Let $a, b > 0$. There is a finite signed Borel measure $\mu_{a,b}$ on $[0, 1]$ satisfying*

$$\int_{[0,1]} u^{n-1} d\mu_{a,b}(u) = m_n(a, b), \quad n \geq 1, \quad (3.3)$$

if and only if $a + b \geq 1$. Whenever it exists, it is unique and has total mass zero. More precisely:

1. *If $a + b > 1$, the measure is $d\mu_{a,b}(u) = \kappa_{a,b}(u) du$, with $\kappa_{a,b} \in L^1(0, 1)$. There is a unique $r_{a,b} \in (0, 1)$ such that*

$$\kappa_{a,b}(u) < 0 \quad (0 < u < r_{a,b}), \quad \kappa_{a,b}(u) > 0 \quad (r_{a,b} < u < 1). \quad (3.4)$$

The zero is simple in the coordinate $T = -\log u$.

2. *If $a + b = 1$, then $0 < a, b < 1$ and*

$$d\mu_{a,b}(u) = \frac{1}{a} d\delta_1(u) + \kappa_{a,b}(u) du. \quad (3.5)$$

The same formula (3.2) defines an integrable density, and

$$\kappa_{a,b}(u) < -1 \quad (0 < u < 1), \quad \int_0^1 \kappa_{a,b}(u) du = -\frac{1}{a}. \quad (3.6)$$

Thus all positive mass is the endpoint atom $1/a$ at $u = 1$.

Proof. We prove convergence, signs, and the boundary atom separately.

Integrability and moments when $a + b > 1$. For $n > 0$, direct integration gives

$$\begin{aligned} \int_0^\infty e^{-nT} [T^{a-1} - (T-v)_+^{a-1}] dT \\ = \Gamma(a)n^{-a}(1 - e^{-nv}). \end{aligned} \quad (3.7)$$

If $a \geq 1$, the bracket is nonnegative. Tonelli's theorem applies, and at $n = 1$ the positive integral contribution in (3.2) is exactly one. The explicit negative term has absolute integral one against $e^{-T} dT$ as well.

If $0 < a < 1$, write

$$B_n(v) = \int_0^\infty e^{-nT} |T^{a-1} - (T-v)_+^{a-1}| dT.$$

For $0 < v \leq 1$, split the T -integral at $2v$. The part before $2v$ is $O(v^a)$. On $T \geq 2v$, the mean value theorem bounds the difference by $C_a v T^{a-2}$, whose integral is also $O(v^a)$. Thus $B_n(v) = O(v^a)$ near zero. For $v \geq 1$, separate gamma integrals give $B_n(v) \leq C_n$. Since $v^{b-1}/(e^v - 1) = O(v^{b-2})$ at zero, the double absolute integral converges if $a + b > 1$. This proves integrability and justifies Fubini in (3.7). For integers $n \geq 1$,

$$\begin{aligned} \int_0^\infty e^{-nT} k_{a,b}(T) dT &= -n^{-a-b} + \frac{n^{-a}}{\Gamma(b)} \int_0^\infty \frac{v^{b-1}(1 - e^{-nv})}{e^v - 1} dv \\ &= -n^{-a-b} + n^{-a} \sum_{j=1}^n j^{-b} = m_n(a, b). \end{aligned} \quad (3.8)$$

The finite geometric identity $(1 - e^{-nv})/(e^v - 1) = \sum_{j=1}^n e^{-jv}$ proves the second line. The change of variable $u = e^{-T}$ gives (3.3); at $n = 1$ it also gives zero total mass.

The crossing problem as a fixed positive level. Define

$$g(s) = s^{b-1} [1 - (1-s)_+^{a-1}], \quad I(T) = \int_0^\infty \frac{g(s)}{e^{Ts} - 1} ds. \quad (3.9)$$

Substitution $v = Ts$ gives

$$\frac{k_{a,b}(T)}{T^{a+b-1}} = -\frac{1}{\Gamma(a+b)} + \frac{I(T)}{\Gamma(a)\Gamma(b)}. \quad (3.10)$$

If $a \geq 1$, then g is nonnegative and positive on a set of positive measure, so $I'(T) < 0$.

If $0 < a < 1$, then $g < 0$ on $(0, 1)$ and $g > 0$ on $(1, \infty)$. For $T_2 > T_1 > 0$ the ratio

$$R(s) = \frac{e^{T_1 s} - 1}{e^{T_2 s} - 1}$$

is strictly decreasing in s and satisfies $0 < R(s) < 1$. Indeed, logarithmic differentiation reduces this assertion to the strict increase of $t/(1 - e^{-t})$ for $t > 0$. If $I(T_1) = C > 0$, then

$$I(T_2) - R(1)I(T_1) = \int_0^\infty \frac{g(s)}{e^{T_1 s} - 1} (R(s) - R(1)) ds < 0.$$

Both parts of the integrand are strictly negative. Consequently $I(T_2) < C$. When $T_2 < T_1$, the ratio is instead strictly increasing and exceeds one, yielding $I(T_2) > C$. Therefore I crosses any fixed positive level at most once.

At such a crossing the derivative is strictly negative. In fact,

$$q_T(s) = -\frac{s}{1 - e^{-Ts}}$$

is strictly decreasing and negative; subtracting $q_T(1)I(T)$ from the differentiated integral gives $I'(T) < q_T(1)I(T) < 0$. All differentiations follow from domination on compact positive T -intervals. Thus a crossing of the positive level $\Gamma(a)\Gamma(b)/\Gamma(a+b)$ in (3.10), if it exists, is unique and simple.

Endpoint signs. Dominated convergence in (3.9) shows that $I(T) \rightarrow 0$ as $T \rightarrow \infty$. Hence $k_{a,b}$ is negative for large T . At the other endpoint one has

$$k_{a,b}(T) \sim \begin{cases} \frac{\zeta(b)}{\Gamma(a)} T^{a-1}, & b > 1, \\ \frac{T^{a-1}}{\Gamma(a)} \log(1/T), & b = 1, \\ \frac{T^{a+b-2}}{(1-b)\Gamma(a+b-1)}, & 0 < b < 1, a+b > 1. \end{cases} \quad (3.11)$$

For the first two cases, divide (3.2) by T^{a-1} and split its integral at $v = T$. The part on $(0, T)$ is $O(T^{b-1})$ after $v = Ts$: cancellation at $s = 0$ and the exponent $a-1 > -1$ at $s = 1$ give a finite constant. The remaining part is $\int_T^\infty v^{b-1}/(e^v - 1) dv$, which tends to $\Gamma(b)\zeta(b)$ for $b > 1$ and equals $\log(1/T) + O(1)$ for $b = 1$.

For $0 < b < 1$, the absolutely convergent beta identity

$$\int_0^\infty \frac{g(s)}{s} ds = \frac{a+b-1}{1-b} B(b, a) \quad (3.12)$$

holds for every $a > 0$. To prove it without analytic continuation, first replace a by $a+1$ and integrate by parts; this gives $aB(b, a)/(1-b)$. On $(0, 1)$ use

$$1 - (1-s)^{a-1} = [1 - (1-s)^a] - s(1-s)^{a-1}$$

and subtract $B(b, a)$. These operations involve only convergent integrals. Since $T/(e^{Ts} - 1) \leq 1/s$, dominated convergence gives

$$TI(T) \rightarrow \frac{a+b-1}{1-b} B(b, a).$$

For $a+b > 1$, inserting this limit in (3.10) proves the third case of (3.11). Thus $k_{a,b}$ is positive near zero. Its unique simple crossing follows, and $u = e^{-T}$ reverses the order of the signs as in (3.4).

The critical density. Suppose $a+b = 1$. In this case $0 < a, b < 1$, and (3.12) says that the integrable function $c(s) = g(s)/s$ has zero integral. It is negative on $(0, 1)$ and positive on $(1, \infty)$. The function

$$q_T(s) = \frac{s}{e^{Ts} - 1}$$

is strictly decreasing. Subtracting $q_T(1) \int c = 0$ therefore gives $I(T) = \int c(s)q_T(s) ds < 0$. Equation (3.10), with $a+b = 1$, proves $k_{a,b}(T) < -1$.

The critical integrability assertion needs cancellation. Use $\int c = 0$ and the elementary bound

$$0 \leq \frac{1}{T} - \frac{s}{e^{Ts} - 1} \leq \min(s, 1/T).$$

On $(0, 1)$ the integral of $s|c(s)|$ is finite, while on $[1, \infty)$ one has $c(s) = s^{b-2}$. Splitting the latter integral at $1/T$ proves $|I(T)| = O(1 + T^{-b})$ as $T \downarrow 0$. Since $b < 1$, this is integrable in T . For $T \geq 1$ the integral has a fixed absolutely integrable majorant. Hence $e^{-T}k_{a,b}(T)$ is integrable.

One cannot apply (3.8) at this boundary: its unweighted double absolute integral can fail to converge. Instead introduce an additional factor T . Splitting at $2v$ as before yields

$$\int_0^\infty T e^{-nT} \left| T^{a-1} - (T-v)_+^{a-1} \right| dT = O(v) \quad (v \downarrow 0). \quad (3.13)$$

The first part is $O(v^{a+1})$. The remaining part is bounded by $C_a v \int_{2v}^\infty T^{a-1} e^{-nT} dT = O(v)$. The resulting double absolute integral converges for every $b > 0$.

Define, now for any real $x > 0$,

$$m(x) = -x^{-a-b} + \frac{x^{-a}}{\Gamma(b)} \int_0^\infty \frac{v^{b-1}(1 - e^{-xv})}{e^v - 1} dv. \quad (3.14)$$

For integer $x = n$, this is $m_n(a, b)$. Differentiating (3.7), with (3.13) supplying Fubini's hypothesis, gives

$$\frac{d}{dx} \int_0^\infty e^{-xT} k_{a,b}(T) dT = m'(x).$$

The difference between these functions is constant. The Laplace integral tends to zero as $x \rightarrow \infty$, whereas elementary sum–integral comparison gives

$$m_n(a, 1-a) = n^{-a} H_{n-1}^{(1-a)} \longrightarrow \frac{1}{a}.$$

Taking this limit along the integers determines the constant:

$$\int_0^\infty e^{-xT} k_{a,1-a}(T) dT = m(x) - \frac{1}{a}. \quad (3.15)$$

This proves the endpoint atom in (3.5). At $x = 1$ it also gives the density mass in (3.6).

Necessity and uniqueness. If $a + b < 1$, then $b < 1$ and

$$m_n(a, b) \sim \frac{n^{1-a-b}}{1-b} \longrightarrow +\infty.$$

Moments of a finite signed measure on $[0, 1]$ are bounded by its total variation, a contradiction. Finally, polynomials are dense in $C[0, 1]$, so equality of all polynomial moments implies equality of finite signed measures. This proves uniqueness and finishes the classification. $\square$

Corollary 3.2 (A completely monotone critical deficit). *For $0 < a < 1$ define*

$$d_a(x) = \frac{1}{a} - x^{-a} [\zeta(1-a) - \zeta(1-a, x)], \quad x > 0. \quad (3.16)$$

Then $d_a(x) - 1/x$ is strictly completely monotone. In particular, for every integer $j \geq 0$ and $x > 0$,

$$(-1)^j d_a^{(j)}(x) > \frac{j!}{x^{j+1}}. \quad (3.17)$$

The sequence $1/a - n^{-a} H_{n-1}^{(1-a)}$, $n \geq 1$, is a positive Hausdorff moment sequence, represented by the density $-\kappa_{a,1-a}(u) > 1$.

Proof. For $b = 1 - a$, the convergent difference integral

$$\zeta(b) - \zeta(b, x) = \frac{1}{\Gamma(b)} \int_0^\infty \frac{v^{b-1}(e^{-v} - e^{-xv})}{1 - e^{-v}} dv$$

identifies (3.14) with $x^{-a}[\zeta(b) - \zeta(b, x)]$. Consequently (3.15) gives

$$d_a(x) = \int_0^\infty e^{-xT} [-k_{a,1-a}(T)] dT.$$

The density is strictly greater than one. Differentiation is justified by the endpoint bounds in the proof of Theorem 3.1. After j derivatives the integral is strictly greater than $\int_0^\infty T^j e^{-xT} dT = j!/x^{j+1}$. Subtracting $1/x$ leaves the positive density $-k - 1$, proving the complete-monotonicity assertion. Integer x gives the moment statement. $\square$

3.2 The entire zero arc in the upper half unit disk

The finite-measure threshold also supports a complete geometric statement. At the critical line, boundary values must be understood analytically: the coefficients in (3.1) tend to $1/a$, so the unit-circle series does not converge in the ordinary sense.

Theorem 3.3 (Unique simple angular zero at every radius). *Let $a, b > 0$ with $a + b \geq 1$. The function $F_{a,b}$ continues holomorphically to $\mathbb{C} \setminus [1, \infty)$ by*

$$F_{a,b}(z) = \int_{[0,1]} \frac{z}{1 - zu} d\mu_{a,b}(u). \quad (3.18)$$

For each $0 < R \leq 1$, exactly one $\theta_{a,b}(R) \in (0, \pi)$ satisfies $\Im F_{a,b}(Re^{i\theta_{a,b}(R)}) = 0$. It is simple, its angular derivative is negative, and

$$\arccos R < \theta_{a,b}(R) < \frac{\pi}{2}. \quad (3.19)$$

The imaginary part is positive before this zero and negative after it as θ increases from 0 to π .

Set $s = R^2$ and $x_{a,b}(s) = R \cos \theta_{a,b}(R)$. Then $x_{a,b}$ is real analytic on $(0, 1]$, extends real analytically through $s = 0$, and

$$0 < x_{a,b}(s) < s, \quad x'_{a,b}(s) > 0 \quad (0 < s \leq 1). \quad (3.20)$$

Proof. Expanding the kernel of (3.18) for $|z| < 1$ and applying Theorem 3.1 proves the identity there. On compact subsets of the slit plane its denominator is bounded away from zero uniformly for $u \in [0, 1]$, so the finite measure proves holomorphy and continuation.

We use an elementary sign test. If a zero-mass signed measure has its negative and positive parts separated, integration against a strictly increasing function is strictly positive, and integration against a strictly decreasing function is strictly negative. Subtract the value of the test function at the separating point to prove this. For $a + b > 1$ use the crossing $r_{a,b}$ from (3.4). For $a + b = 1$ use the separating point 1: the negative density lies in $(0, 1)$ and the positive mass is at 1. Strictness still follows on the interval occupied by the density.

Write $c = \cos \theta$ and

$$K_c(u) = \frac{1}{1 - 2Rcu + R^2u^2}, \quad J_R(c) = \int_{[0,1]} K_c(u) d\mu_{a,b}(u).$$

Then

$$\Im F_{a,b}(Re^{i\theta}) = R \sin \theta J_R(\cos \theta). \quad (3.21)$$

The kernel K_0 is strictly decreasing in u , so $J_R(0) < 0$. For $R < 1$, an absolutely convergent geometric expansion gives

$$J_R(1) = \sum_{k \geq 0} (k+1) R^k m_{k+1}(a, b) \geq 2Rm_2(a, b) > 0. \quad (3.22)$$

For $R = 1$ and $a + b > 1$, split the measure at its sign crossing. The negative part stays away from $u = 1$, so its contribution has a finite limit; monotone convergence applies to the positive part. Passing $R \uparrow 1$ in (3.22) shows that the signed integral against $(1-u)^{-2}$ is at least $2m_2 > 0$, possibly infinite. Passing $c \uparrow 1$ in $J_1(c)$ gives the same integral by the same separation.

At $a + b = 1$, the negative part reaches the endpoint, so that split argument must be replaced. Since

$$1 - 2cu + u^2 = (u - c)^2 + 1 - c^2, \quad 0 \leq \frac{1 - c}{1 - 2cu + u^2} \leq \frac{1}{1 + c} \leq 1$$

for $0 \leq c < 1$, dominated convergence against the finite signed measure gives

$$\lim_{c \uparrow 1} (1 - c) J_1(c) = \frac{1}{2} \mu_{a,b}(\{1\}) = \frac{1}{2a} > 0. \quad (3.23)$$

The pointwise multiplier tends to zero at every $u < 1$ and to $1/2$ at $u = 1$. Thus $J_1(c) \rightarrow +\infty$. In all cases continuity produces at least one zero $c_0 \in (0, 1)$.

For $c' > c$ one has

$$\frac{d}{du} \frac{K_{c'}(u)}{K_c(u)} = \frac{2R(c' - c)(1 - R^2 u^2)}{(1 - 2Rc'u + R^2 u^2)^2} > 0 \quad (0 < u < 1). \quad (3.24)$$

At a zero $J_R(c) = 0$, the signed measure $K_c d\mu$ has zero mass and the same separated signs. Applying the sign test to (3.24) gives $J_R(c') > 0$ for $c' > c$ and $J_R(c') < 0$ for $c' < c$. This proves uniqueness on $(-1, 1)$ and the signs asserted in the theorem.

The logarithmic derivative of K_c with respect to c is $2Ru/(1 - 2Rcu + R^2 u^2)$, a strictly increasing function of u . The sign test therefore gives $J'_R(c_0) > 0$. At the corresponding angular zero, (3.21) has derivative $-R \sin^2 \theta J'_R(c_0) < 0$.

To sharpen the enclosure, put $H(u) = -\mu_{a,b}([0, u])$ for $0 < u < 1$. The sign structure and zero total mass imply $H(u) > 0$ throughout this interval. Integration by parts for functions of bounded variation, including the endpoint jump when $a + b = 1$, gives

$$J_R(c) = 2R \int_0^1 \frac{H(u)(c - Ru)}{(1 - 2Rcu + R^2 u^2)^2} du. \quad (3.25)$$

At a zero, c/R is therefore a strictly interior weighted mean of $u \in (0, 1)$, proving $0 < c < R$ and (3.19).

For the motion of the abscissa, treat x and s as independent variables and write

$$G(x, s) = \int_{[0,1]} \frac{d\mu(u)}{D(u)}, \quad D(u) = 1 - 2xu + su^2.$$

At a zero let $d\sigma = d\mu/D$. This measure again has zero mass and separated signs. Since $0 < x < s \leq 1$,

$$\begin{aligned} G_x &= 2 \int \frac{u}{D(u)} d\sigma(u) > 0, & \left(\frac{u}{D}\right)' &= \frac{1 - su^2}{D^2} > 0, \\ G_s &= - \int \frac{u^2}{D(u)} d\sigma(u) < 0, & \left(\frac{u^2}{D}\right)' &= \frac{2u(1 - xu)}{D^2} > 0 \end{aligned}$$

on $0 < u < 1$. The implicit-function theorem yields $x'(s) = -G_s/G_x > 0$. At $(x, s) = (0, 0)$ one has $G = 0$ and $G_x = 2m_2 > 0$, giving an analytic extension through $s = 0$. At $s = 1$ the root is away from the cut, so the same argument applies in a neighborhood. This proves (3.20) and the analyticity assertions. $\square$

Corollary 3.4 (Real-weight Gaussian negativity). *For every $a, b > 0$ with $a + b \geq 1$,*

$$\Im F_{a,b}(i) < 0.$$

On the critical line this is a statement about the analytic, equivalently Abel, boundary value.

Proof. Apply Theorem 3.3 at $R = 1$ and $\theta = \pi/2$, which lies strictly after the unique angular zero. $\square$

Proposition 3.5 (Exact small-radius coefficients). *Suppress the parameters in m_n and put*

$$\beta = \frac{m_3}{2m_2} = \frac{H_2^{(b)}}{2} \left(\frac{2}{3}\right)^a, \quad \eta = \frac{4m_4\beta - 4m_3\beta^2 - m_5}{2m_2}.$$

As $s \rightarrow 0$ and $R \rightarrow 0$,

$$\begin{aligned} x_{a,b}(s) &= \beta s + \eta s^2 + O(s^3), \\ \frac{\pi}{2} - \theta_{a,b}(R) &= \beta R + \left(\eta + \frac{\beta^3}{6}\right) R^3 + O(R^5). \end{aligned}$$

Every further coefficient is obtained recursively from the moments.

Proof. The analytic equation for G gives

$$G(x, s) = 2m_2x - m_3s + 4m_3x^2 - 4m_4xs + m_5s^2 + O((|x| + |s|)^3).$$

Substitute $x = \beta s + \eta s^2 + O(s^3)$ and compare coefficients. The second formula follows from $\pi/2 - \theta(R) = \arcsin(x(R^2)/R)$. At every further order the new coefficient is multiplied by $2m_2 > 0$, giving a triangular recursion. $\square$

Example 3.6 (An exact circular arc). For $a = b = 1$, the elementary identity $F_{1,1}(z) = \frac{1}{2} \log^2(1 - z)$ shows that the nonreal zero arc of the imaginary part in the unit disk is $|1 - z| = 1$. Indeed, there $\arg(1 - z)$ is strictly negative, so the imaginary part of the squared logarithm vanishes exactly when $\log|1 - z| = 0$. Thus

$$x_{1,1}(s) = \frac{s}{2}, \quad \theta_{1,1}(R) = \arccos(R/2), \quad \theta_{1,1}(1) = \frac{\pi}{3}.$$

The density itself is $k_{1,1}(T) = -\log(e^T - 1)$, with crossing $T = \log 2$ and $r = 1/2$.

Corollary 3.7 (Continuous outer-order motion of the density crossing). *Fix $b > 0$. If $a_2 > a_1 > 0$ and $a_1 + b > 1$, then $r_{a_2,b} < r_{a_1,b}$.*

Proof. Put $\delta = a_2 - a_1 > 0$. Multiplication of the moments by $n^{-\delta}$ and uniqueness in Theorem 3.1 give the Mellin-convolution formula

$$k_{a_2,b}(T) = \frac{1}{\Gamma(\delta)} \int_0^T (T - v)^{\delta-1} k_{a_1,b}(v) dv.$$

Absolute exponential integrability justifies convolution and its Laplace transform. Both sides are continuous for $T > 0$, so moment uniqueness gives pointwise equality. At the old logarithmic crossing $T_1 = -\log r_{a_1,b}$ the integrand is positive almost everywhere. The new density is therefore positive at T_1 , and its unique crossing occurs at $T_2 > T_1$. Returning to $r = e^{-T}$ proves the claim. $\square$

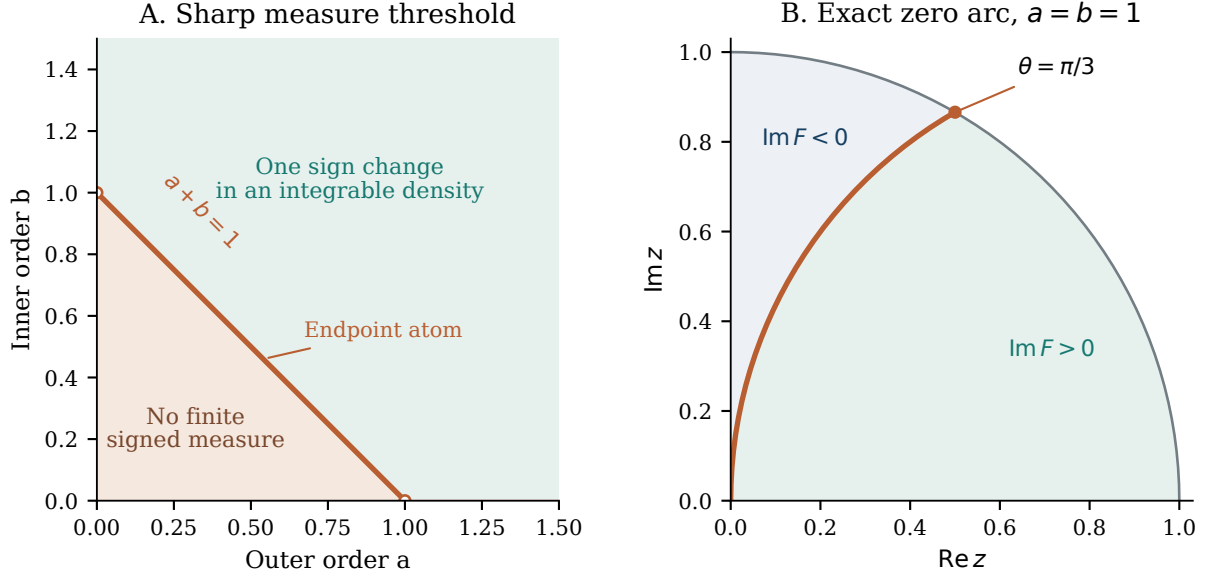


Figure 1: The sharp measure domain for positive real orders (left), and an exact illustration of the angular-zero theorem in the first quadrant at $a = b = 1$ (right). The separating arc is $|1 - z| = 1$ and meets the unit circle at angle $\pi/3$. The phase diagram is a theorem, and the circular arc is evaluated exactly.

3.3 Verification and the limits of the geometric conclusion

The diagnostic program `code/geometry_diagnostics.py` and its output `data/geometry_diagnostics.json` accompany these proofs. They compare two kernel quadratures at nine points; locate four density crossings, including $(a, b) = (0.5005, 0.5)$ near the critical line; check fifteen strictly negative critical-density values; and compute sixteen interior-disk zero locations. The exact arc in Example 3.6 supplies an independent comparator. Twelve additional samples check the critical deficit and its first four signed derivatives. All diagnostic assertions pass. The checks are numerical and are not interval certificates; the existence, uniqueness, and sign assertions rest on the analytic arguments above.

Near the critical line, direct subtraction of the two pieces of the kernel can be unstable. For $0 < a, b < 1$ the evaluator instead uses

$$I(T) = q_T(1) \int_0^\infty c(s) ds + \int_0^\infty c(s) [q_T(s) - q_T(1)] ds, \quad c(s) = g(s)/s.$$

The first integral is the exact beta expression (3.12); the second integrand is negative on both sides of $s = 1$. At $a + b = 1$ the beta term vanishes exactly. This is the cancellation used in the proof, translated into a stable evaluator.

Three distinctions delimit the result. First, $x'_{a,b}(R^2) > 0$ does not by itself imply $\theta'_{a,b}(R) < 0$; that stronger angular monotonicity is not proved here. Second, the monotone motion of the *density* crossing with a does not imply monotone motion of the angular zero with a or b . Third, when $a + b < 1$ the finite signed-measure method is impossible by Theorem 3.1, but this does not settle zero uniqueness by other methods. The next section quantifies concentration into the endpoint atom for fixed $b \in (0, 1)$, providing a scale for uniform numerical evaluation.

4 The critical boundary: a power-law crossing and an exponential mass law

The atom in Theorem 3.1 raises a quantitative question: how does an absolutely continuous positive part collapse into that atom? Two distinct scales occur. Its outermost point approaches the endpoint by a power of the distance to the critical line. Most of its mass lies much closer to the endpoint, on an exponential scale described by a limiting exponential distribution.

Fix $b \in (0, 1)$, and write

$$p = 1 - b, \quad a = p + \varepsilon, \quad k_\varepsilon(T) = k_{p+\varepsilon,b}(T), \quad 0 < \varepsilon \leq \varepsilon_0 < b. \quad (4.1)$$

Let T_ε be the unique logarithmic crossing of k_ε , so $k_\varepsilon > 0$ on $(0, T_\varepsilon)$. All constants below may depend on the fixed b ; the local expansion constants may also depend on its chosen radius T_0 .

Lemma 4.1 (A uniform local kernel expansion). *For any fixed $T_0 < 2\pi$, uniformly for $0 \leq \varepsilon \leq \varepsilon_0$ and $0 < T \leq T_0$,*

$$k_\varepsilon(T) = T^{\varepsilon-1} \left\{ \frac{\varepsilon}{p\Gamma(1+\varepsilon)} + \frac{\zeta(b)}{\Gamma(p+\varepsilon)} T^p - \frac{T}{2\Gamma(1+\varepsilon)} - \frac{bT^2}{12\Gamma(2+\varepsilon)} + O_{b,T_0}(T^4) \right\}. \quad (4.2)$$

More generally the bracket has the convergent expansion

$$\begin{aligned} & \frac{\varepsilon}{p\Gamma(1+\varepsilon)} + \frac{\zeta(b)}{\Gamma(p+\varepsilon)} T^p - \frac{T}{2\Gamma(1+\varepsilon)} \\ & - \sum_{\ell \geq 1} \frac{B_{2\ell}\Gamma(b+2\ell-1)}{(2\ell)!\Gamma(b)\Gamma(\varepsilon+2\ell)} T^{2\ell}, \quad 0 < T < 2\pi. \end{aligned} \quad (4.3)$$

Here $B_{2\ell}$ are the Bernoulli numbers, with $B_2 = 1/6$.

Proof. Retain g, I from (3.9), now with $a = p + \varepsilon$, and set $h(v) = (e^v - 1)^{-1} - v^{-1}$. Subtracting the leading singularity inside the absolutely convergent integral gives the exact identity

$$\begin{aligned} I(T) &= \frac{J}{T} + \Gamma(b)\zeta(b)T^{-b} \\ & - \int_0^1 s^{b-1}(1-s)^{a-1}h(Ts)ds, \quad J = \frac{\varepsilon}{p}B(b, a). \end{aligned} \quad (4.4)$$

Indeed, $J = \int g(s)/s ds$ by (3.12), and

$$\int_0^\infty s^{b-1}h(Ts)ds = T^{-b} \int_0^\infty v^{b-1}h(v)dv = \Gamma(b)\zeta(b)T^{-b}.$$

The last integral converges for $0 < b < 1$. To identify it, subtract $1/v$ from the Bose integral only on $(0, 1)$ in the half-plane $\Re b > 1$ and continue the resulting expression to $\Re b > 0$, $b \neq 1$. The added term is $1/(b-1)$. For $0 < b < 1$ it is exactly $-\int_1^\infty v^{b-2}dv$, giving the displayed integral. In particular $\zeta(b) < 0$, since $h(v) < 0$ for $v > 0$.

Only a compact beta integral remains in (4.4). The convergent Taylor expansion

$$h(v) = -\frac{1}{2} + \frac{v}{12} - \frac{v^3}{720} + \cdots, \quad |v| < 2\pi,$$

may be integrated termwise there. For uniformity, choose a number T_1 with $T_0 < T_1 < 2\pi$. Cauchy's estimates on $|v| = T_1$ bound the Taylor remainders uniformly for $|v| \leq T_0$; $s^{b-1}(1-s)^{p-1}$ dominates the beta weights for $0 \leq \varepsilon \leq \varepsilon_0$. Substituting the integrated series into (3.10) proves (4.3) and its stated uniform truncation. $\square$

Theorem 4.2 (Crossing scale and its first correction). *Define*

$$\tau_\varepsilon = \left\{ \frac{\Gamma(p+\varepsilon)\varepsilon}{p[-\zeta(b)]\Gamma(1+\varepsilon)} \right\}^{1/p}. \quad (4.5)$$

As $\varepsilon \downarrow 0$,

$$T_\varepsilon = \tau_\varepsilon - \frac{\tau_\varepsilon^2}{2\varepsilon} + O_b\left(\frac{\tau_\varepsilon^3}{\varepsilon^2}\right). \quad (4.6)$$

In particular, with

$$K_b = \left\{ \frac{\Gamma(1-b)}{(1-b)[- \zeta(b)]} \right\}^{1/(1-b)}, \quad (4.7)$$

one has

$$T_\varepsilon \sim K_b \varepsilon^{1/(1-b)}, \quad 1 - r_{p+\varepsilon, b} \sim K_b \varepsilon^{1/(1-b)}. \quad (4.8)$$

Proof. For every fixed $T > 0$, the kernel depends continuously on ε and its limit $k_{p,b}(T)$ is strictly negative by (3.6). The unique crossing therefore satisfies $T_\varepsilon \rightarrow 0$. At that crossing, multiply the bracket in (4.2) by $\Gamma(1+\varepsilon)$ and write

$$C_\varepsilon = \frac{[-\zeta(b)]\Gamma(1+\varepsilon)}{\Gamma(p+\varepsilon)} > 0.$$

This gives

$$\frac{\varepsilon}{p} = C_\varepsilon T_\varepsilon^p + \frac{T_\varepsilon}{2} + \frac{bT_\varepsilon^2}{12(1+\varepsilon)} + O_b(T_\varepsilon^4). \quad (4.9)$$

Since $p < 1$, the last three terms are lower order than T_ε^p . Hence $T_\varepsilon/\tau_\varepsilon \rightarrow 1$.

Put $y = T_\varepsilon/\tau_\varepsilon$ and $q = \tau_\varepsilon/\varepsilon$. The leading equivalence implies $q = O_b(\varepsilon^{b/p}) \rightarrow 0$. Dividing (4.9) by ε/p gives, uniformly for y near one,

$$1 = y^p + \frac{p}{2}qy + O_b(\varepsilon q^2).$$

The mean value theorem first yields $y - 1 = O_b(q)$. Taylor expansion then yields $p(y - 1) + (p/2)q = O_b(q^2)$, so $y = 1 - q/2 + O_b(q^2)$. This proves (4.6). Formula (4.8) follows from continuity of the gamma factors and $1 - e^{-T} \sim T$. $\square$

At the symmetric boundary point $b = p = 1/2$, the analytic correction from the gamma quotient and the nonlinear root correction have the same relative order. Using $\psi(1/2) = -\gamma - 2 \log 2$ gives the explicit consequence

$$T_\varepsilon = \frac{4\pi}{\zeta(1/2)^2} \varepsilon^2 \left[1 - \left(4 \log 2 + \frac{2\pi}{\zeta(1/2)^2} \right) \varepsilon + O(\varepsilon^2) \right]. \quad (4.10)$$

For other b , retaining the exact gamma quotient in τ_ε keeps the algebraic and fractional-power corrections in (4.6) unambiguous.

Theorem 4.3 (Condensation and the exponential positive-mass law). *Let $\mu_\varepsilon = \mu_{p+\varepsilon,b}$, and denote its positive and negative parts by μ_ε^+ and μ_ε^- , so $\mu_\varepsilon = \mu_\varepsilon^+ - \mu_\varepsilon^-$. Then*

$$\mu_\varepsilon^- \longrightarrow -\kappa_{p,b}(u) du \quad \text{in total variation}, \quad \mu_\varepsilon^+ \Longrightarrow \frac{1}{p}\delta_1. \quad (4.11)$$

In particular $\mu_\varepsilon \Longrightarrow \mu_{p,b}$.

Let ν_ε be the pushforward of the finite positive measure $p k_\varepsilon(T)^+ e^{-T} dT$ under $Y_\varepsilon = -\varepsilon \log T$. For all sufficiently small ε it is supported in $[0, \infty)$ and

$$\|\nu_\varepsilon - e^{-y} \mathbf{1}_{y>0} dy\|_{TV} = O_b(\varepsilon |\log \varepsilon|). \quad (4.12)$$

The same convergence and rate hold after normalizing ν_ε to a probability measure.

Proof. For $T \leq T_0 < 1$, Lemma 4.1 gives

$$k_\varepsilon(T) = \frac{\varepsilon}{p\Gamma(1+\varepsilon)} T^{\varepsilon-1} + R_\varepsilon(T), \quad |R_\varepsilon(T)| \leq C_b T^{p+\varepsilon-1}. \quad (4.13)$$

The first term is positive, so the negative part is bounded by $C_b T^{p-1}$, an integrable bound independent of ε . For $T \geq T_0$, the scaled integral (3.9) has a common integrable majorant as a ranges over $[p, p + \varepsilon_0]$; hence $e^{-T} |k_\varepsilon(T)| \leq C_b e^{-T} (1 + T^{\varepsilon_0})$. Pointwise convergence and dominated convergence prove the negative-part assertion in (4.11). Its limiting mass is $1/p$ by Theorem 3.1. Each μ_ε has total mass zero, so the positive masses have the same limit. Their supports lie in $(e^{-T_\varepsilon}, 1)$, which shrinks to 1; this proves the positive weak limit.

For the quantitative distributional statement set $y_\varepsilon = -\varepsilon \log T_\varepsilon$. Theorem 4.2 gives $y_\varepsilon = O_b(\varepsilon |\log \varepsilon|)$ and $y_\varepsilon > 0$ for small ε . On the positive support $0 < T < T_\varepsilon$, replace the density in (4.13) by its leading term and replace e^{-T} by one. The total variation error is at most

$$C_b T_\varepsilon^{p+\varepsilon} + C_b \varepsilon T_\varepsilon^{1+\varepsilon} = O_b(\varepsilon),$$

because $T_\varepsilon^p = O_b(\varepsilon)$. Total variation cannot increase under pushforward. With $T = e^{-y/\varepsilon}$, the pushforward of the resulting leading measure is exactly

$$\frac{e^{-y}}{\Gamma(1+\varepsilon)} \mathbf{1}_{y>y_\varepsilon} dy.$$

Its total variation difference from the unit exponential density is at most $O(\varepsilon) + \int_0^{y_\varepsilon} e^{-y} dy$. This proves (4.12). Its mass consequently differs from one by the same order; normalizing changes the total variation by at most that mass discrepancy. $\square$

Thus the power-law crossing width does not describe the typical positive-mass location. Under the normalized positive part, $-\varepsilon \log(-\log u)$ tends to an exponential random variable of mean one. A typical logarithmic distance $T = -\log u$ is therefore exponentially small in $1/\varepsilon$, even though the outer edge of the positive region is of order $\varepsilon^{1/(1-b)}$.

The accompanying boundary diagnostics compare the two-term root formula at $b = 1/4, 1/2, 3/4$ and decreasing ε , and compare positive-mass distribution functions with the exponential law. The convergent series (4.3) provides a stable evaluator with a recorded analytic truncation bound. Numerical rounding is still separate from that bound; these are diagnostics, not new proof assumptions.

All limits in this section keep b fixed. As $b \downarrow 0$, the small parameter $\tau_\varepsilon/\varepsilon$ has exponent $b/(1-b) \downarrow 0$, so the stated hierarchy is not uniform at that endpoint. As $b \uparrow 1$, the exponent $1/(1-b)$ and the constants become singular. Joint limits involving either endpoint require additional analysis and are not consequences of these theorems.

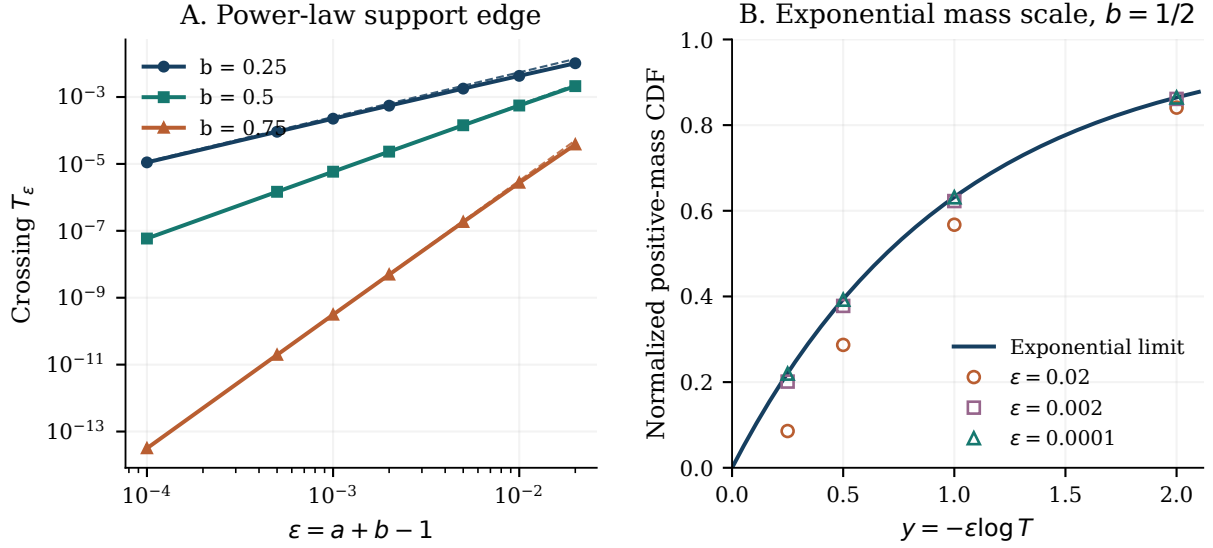


Figure 2: The two scales of endpoint concentration. Left: numerical crossings (markers) and the leading power laws (dashed curves) at three fixed inner orders. Right: four sampled distribution-function values for each indicated ϵ at $b = 1/2$, compared with the proved unit-exponential limit. These are diagnostics of the analytic results.

5 A joint transition for reflected log-gamma moments

The reflected family already considered in the manuscript is

$$M_{n,m} = \int_0^1 f(x)^n f(1-x)^m dx, \quad f(x) = \log \Gamma(x), \quad n, m \in \mathbb{Z}_{\geq 0}. \quad (5.1)$$

These mixed moments occur in Amdeberhan et al. [5, Section 5]. The manuscript proves that, for each fixed m ,

$$M_{n,m} \sim \frac{\gamma^m n!}{(m+1)^{n+1}}, \quad n \rightarrow \infty, \quad (5.2)$$

and develops the associated residues, late coefficients, and optimal truncation. Its reflected-moment section explicitly leaves uniformity for growing m open. The result below identifies a transition where the leading expression (5.2) acquires a nontrivial factor. It concerns simultaneous growth of the two exponents, rather than the late truncation index in the already established fixed- m expansion.

5.1 The critical scale and the first correction

Write γ for Euler's constant, and define

$$c = \frac{\zeta(2)}{2\gamma} - \gamma, \quad d = \frac{\zeta(3)}{3\gamma} - \frac{\zeta(2)^2}{8\gamma^2}. \quad (5.3)$$

The constant c is positive; for example $0 < \gamma < 0.6$ and $\zeta(2) > 1$ suffice to prove this. Its numerical value is $c = 0.8476712247576688767895678092 \dots$. For $m \geq 1$ put

$$t = \frac{n}{m+1}, \quad \lambda = me^{-t}, \quad \mathcal{R}_{n,m} = \frac{(m+1)^{n+1} M_{n,m}}{\gamma^m n!}. \quad (5.4)$$

All limits below are along integer pairs (n, m) satisfying the stated conditions. In particular $n > 0$ and $t \geq 2$ for all sufficiently large m .

Theorem 5.1 (Uniform reflected-moment transition). *Fix $S > 0$. Uniformly as $m \rightarrow \infty$ in the strip $|t - \log m| \leq S$, one has*

$$\boxed{\mathcal{R}_{n,m} = e^{c\lambda} \left(1 + \frac{C_1(t, \lambda)}{m} + O_S \left(\frac{t^2}{m^2} \right) \right)}, \quad (5.5)$$

where the correction is the explicit polynomial

$$C_1(t, \lambda) = \lambda \left[c \left(\frac{t}{2} - 1 \right) - 2\gamma \right] + \lambda^2 \left[\frac{c^2 t}{2} + \gamma^2 + d \right]. \quad (5.6)$$

Consequently, if $n/(m+1) - \log m \rightarrow s \in \mathbb{R}$, then

$$\mathcal{R}_{n,m} \rightarrow \exp(ce^{-s}). \quad (5.7)$$

For example, choosing n to be the nearest integer to $(m+1) \log m$ gives $\mathcal{R}_{n,m} \rightarrow e^c = 2.3342\dots$, rather than 1. Thus the fixed-reflected-exponent leading term cannot be used uniformly through this transition. The exact exponential in (5.7) is more informative than a first-order correction to a fixed- m formula.

5.2 The exact probability representation

Set

$$B(x) = \log \Gamma(1-x), \quad \ell(x) = \log \Gamma(1+x).$$

The standard expansions needed below are convergent for $|x| < 1$ [8, Section 5.7]:

$$\begin{aligned} \ell(x) &= -\gamma x + \frac{\zeta(2)}{2} x^2 - \frac{\zeta(3)}{3} x^3 + \dots, \\ B(x) &= \gamma x + \frac{\zeta(2)}{2} x^2 + \frac{\zeta(3)}{3} x^3 + \dots. \end{aligned} \quad (5.8)$$

The inequality $\psi'(x) > 0$, together with $\psi(1) = -\gamma < 0$, shows that $f(x) > 0$ on $(0, 1)$. Log-convexity of Γ between 1 and 2 gives $\ell(x) \leq 0$, hence

$$0 < f(x) \leq -\log x \quad (0 < x < 1), \quad \int_0^1 f(x)^m dx \leq m!. \quad (5.9)$$

Let $r = m+1$ and let Y have the gamma density with shape $n+1$ and rate r :

$$p(y) = \frac{r^{n+1}}{n!} y^n e^{-ry}, \quad y > 0.$$

Its mode is exactly $t = n/r$. The substitution $x = e^{-y}$ in (5.1) gives the identity

$$\mathcal{R}_{n,m} = \mathbb{E} e^{E_m(Y)}, \quad (5.10)$$

where

$$E_m(y) = n \log \left(1 + \frac{\ell(e^{-y})}{y} \right) + m \log \left(\frac{B(e^{-y})}{\gamma e^{-y}} \right). \quad (5.11)$$

The expressions inside both logarithms are positive for $y > 0$. Equation (5.10) is exact, with no asymptotic interchange.

5.3 Why the weighted tails are negligible

Ordinary gamma concentration is not by itself sufficient in (5.10): the multiplying weight can grow on the left. We therefore prove the required estimate explicitly.

Lemma 5.2 (Uniform weighted localization). *For each fixed $S > 0$, there are positive constants C_S, c_S such that, for all sufficiently large m with $|t - \log m| \leq S$,*

$$\mathbb{E} \left[e^{E_m(Y)} \mathbf{1}_{\{|Y-t|>1\}} \right] \leq C_S \exp \left(-c_S \frac{m}{t} \right). \quad (5.12)$$

Unweighted tails with any fixed polynomial factor in $|Y - t|$ obey the same type of bound, after altering the constants.

Proof. Let $y_0 = \log 2$. For $0 < x \leq 1/2$, positivity of the coefficients of B and $\zeta(j)/j \leq \zeta(2)/2$ for $j \geq 2$ give

$$\frac{B(x)}{\gamma x} \leq 1 + \frac{\zeta(2)x}{2\gamma(1-x)} \leq 1 + Cx, \quad C = \frac{\zeta(2)}{\gamma}.$$

Since $\ell \leq 0$, this implies

$$e^{E_m(y)} \leq \exp(Cme^{-y}), \quad y \geq y_0. \quad (5.13)$$

On $0 < y < y_0$, use the original moment integral and (5.9). The unnormalized contribution is at most $y_0^n m!$. Stirling's bounds show that the logarithm of its normalized contribution is

$$-n \log t + O_S(n + m \log m).$$

In the strip, $m \log m = O_S(n)$, so this is at most $-\frac{1}{2}n \log t$ for sufficiently large m .

On $y_0 \leq y \leq t/2$, the weight is at most $e^{Cm/2}$. The gamma lower-tail bound at half the mode is $e^{-c_0 n}$ for a fixed $c_0 > 0$ once n is large. This follows directly from the moment-generating function $(1 - u/r)^{-n-1}$ and exponential Markov inequality. Since $n/m \sim \log m$, the weighted contribution is $O(e^{-c_1 m t})$.

On $t/2 \leq y \leq t - 1$, write $u = t - y$. The exact ratio to the density at the mode satisfies

$$\frac{p(t-u)}{p(t)} = \exp\{n \log(1 - u/t) + ru\} \leq \exp\left(-\frac{ru^2}{2t}\right). \quad (5.14)$$

The weight in (5.13) is at most $\exp(C\lambda e^u)$. For $1 \leq u \leq t/2$, the maximum of e^u/u^2 occurs at an endpoint. As $\lambda \in [e^{-S}, e^S]$ and $t = \log m + O_S(1)$, it follows that

$$C\lambda e^u \leq \frac{ru^2}{4t}$$

uniformly on this interval when m is large. Multiplication by (5.14), followed by integration and $p(t) = O(\sqrt{r/t})$, gives $O_S(e^{-c_2 m/t})$.

Finally, on $y \geq t + 1$, the weight is bounded by $\exp(C\lambda/e)$. The gamma upper-tail bound for a displacement of one from the mode is $O_S(e^{-c_3 m/t})$. For completeness, with mean $(n+1)/r = t+1/r$, the usual gamma exponent at $t+1$ is $(n+1)(\delta - \log(1+\delta))$, where $\delta = (r-1)/(n+1) \asymp 1/t$; this exponent is bounded below by a positive constant times r/t . The four estimates prove (5.12). Inserting a fixed polynomial in the tails only introduces polynomial factors, which can be absorbed by reducing the positive exponential constant. The density ratio or the same exponential Markov argument gives this last assertion. $\square$

5.4 Proof of the transition and its explicit correction

Proof of Theorem 5.1. For the calculation abbreviate

$$a = -\gamma, \quad a_2 = \zeta(2)/2, \quad b = \zeta(2)/(2\gamma), \quad c = a + b.$$

Let $V = Y - t$. Uniformly for $|v| \leq 1$, set $x = (\lambda/m)e^{-v}$ and use (5.8) with its analytic remainder. One obtains

$$E_m(t+v) = A(v) + \frac{B_1(v)}{m} + O_S(m^{-2}), \quad (5.15)$$

where

$$\begin{aligned} A(v) &= \lambda e^{-v} \left(b + \frac{at}{t+v} \right), \\ B_1(v) &= \frac{a\lambda t e^{-v}}{t+v} + \lambda^2 e^{-2v} \left(d + \frac{a_2 t}{t+v} - \frac{a^2 t}{2(t+v)^2} \right). \end{aligned} \quad (5.16)$$

All derivatives of each fixed order used below are uniformly bounded in this interval; $t/(t+v)$ is bounded once $t \geq 2$.

The exact moments about the mode, rather than about the mean, are

$$\begin{aligned} \mathbb{E}V &= r^{-1}, & \mathbb{E}V^2 &= t/r + 2/r^2, \\ \mathbb{E}V^3 &= 5t/r^2 + 6/r^3, & \mathbb{E}V^4 &= 3t^2/r^2 + 26t/r^3 + 24/r^4. \end{aligned} \quad (5.17)$$

They follow either from gamma raw moments or from $\mathbb{E}e^{zV} = e^{-tz}(1 - z/r)^{-rt-1}$. Taylor expand $e^{A(v)}$ through degree three and bound its fourth remainder with (5.17). The cubic term is $O_S(t/m^2)$ because its signed moment, not its absolute third moment, is used. Expand $e^{A(v)}B_1(v)$ through degree one with a quadratic remainder. Lemma 5.2 allows us to use full gamma moments and discard the outer tails. The result is

$$e^{-A(0)}\mathcal{R}_{n,m} = 1 + \frac{1}{m} \left[B_1(0) + A'(0) + \frac{t}{2} \{A''(0) + A'(0)^2\} \right] + O_S(t^2/m^2). \quad (5.18)$$

Here replacing r^{-1} by m^{-1} changes only the stated remainder.

The local values are

$$\begin{aligned} A(0) &= c\lambda, & A'(0) &= -\lambda(c + a/t), \\ A''(0) &= \lambda(c + 2a/t + 2a/t^2), & B_1(0) &= a\lambda + \lambda^2(d + a_2 - a^2/(2t)). \end{aligned}$$

Substitution into (5.18) cancels all negative powers of t and yields

$$\lambda \{c(t/2 - 1) + 2a\} + \lambda^2 \{c^2 t/2 + ca + d + a_2\}.$$

The relation $ca + a_2 = \gamma^2$ gives (5.6). Since $t/m \rightarrow 0$, the limit (5.7) follows immediately. $\square$

5.5 All fixed asymptotic orders

Theorem 5.3 (Constructive expansion to every fixed order). *For each integer $J \geq 0$, there are explicitly computable functions $C_j(t, \lambda)$, polynomial in λ and rational in t , such that $C_0 = 1$, C_1 is (5.6), and*

$$\mathcal{R}_{n,m} = e^{c\lambda} \left[\sum_{j=0}^J \frac{C_j(t, \lambda)}{m^j} + O_{S,J} \left(\frac{t^{J+1}}{m^{J+1}} \right) \right] \quad (5.19)$$

uniformly in $|t - \log m| \leq S$. For each fixed j , $C_j(t, \lambda) = O_{S,j}(t^j)$. The theorem asserts an expansion at every fixed order, without asserting convergence of the infinite expansion.

The following finite differential prescription makes “computable” precise. Let ϵ be a formal variable and put $r = (1 + \epsilon)/\epsilon$. Define

$$\begin{aligned} \Phi_\epsilon(v) = \epsilon^{-1} & \left[(1 + \epsilon)t \log \left(1 + \frac{\ell(\lambda\epsilon e^{-v})}{t + v} \right) \right. \\ & \left. + \log \left(\frac{B(\lambda\epsilon e^{-v})}{\gamma\lambda\epsilon e^{-v}} \right) \right], \end{aligned} \quad (5.20)$$

$$K_\epsilon(z) = \frac{z}{r} + \sum_{k \geq 2} \left(\frac{t}{kr^{k-1}} + \frac{1}{kr^k} \right) z^k. \quad (5.21)$$

The quotient involving B has constant term one, and both terms in brackets vanish at $\epsilon = 0$, so the formal expression is well defined. With derivatives acting on v , the coefficients are

$$C_j(t, \lambda) = e^{-c\lambda} [\epsilon^j] \exp(K_\epsilon(\partial_v)) e^{\Phi_\epsilon(v)} \Big|_{v=0}. \quad (5.22)$$

Only finitely many derivatives occur at a fixed order: the maximum derivative order at ϵ^j is $2j$.

Proof of Theorem 5.3. Expand (5.11) through order J in $1/m$ on $|v| \leq 1$. The convergent endpoint series make its remainder, together with any fixed number of v derivatives, uniformly $O_{S,J}(m^{-J-1})$. Expand the resulting analytic amplitudes in v through degree $2J + 1$. Gamma moment–cumulant identities give

$$\mathbb{E}|V|^{2J+2} = O_J((t/m)^{J+1}),$$

while signed odd moments occur at integer powers of $1/m$ and have the corresponding smaller order. These facts follow directly by expanding the exact logarithm of the moment-generating function, which is (5.21). Lemma 5.2 controls the tails, including every polynomial moment used here. Taylor’s theorem therefore gives (5.19) with the displayed remainder.

At each fixed order all derivatives at $v = 0$ of $t/(t + v)$ are rational functions of t , and all endpoint coefficients are finite combinations of γ and zeta values. The common exponential factor is $e^{c\lambda}$. This proves the stated coefficient class and (5.22). At formal order j , the moment–cumulant formula introduces at most j positive powers of t ; the remaining amplitude derivatives are bounded as $t \rightarrow \infty$. Consequently $C_j = O_{S,j}(t^j)$. $\square$

For a compact reproducible second-order formula, retain $a = -\gamma$, $a_2 = \zeta(2)/2$ and let $\alpha_j = A^{(j)}(0)$ and $\beta_j = B_1^{(j)}(0)$, using (5.16). Put

$$a_3 = -\zeta(3)/3, \quad e = \frac{\zeta(4)}{4\gamma} - \frac{\zeta(2)\zeta(3)}{6\gamma^2} + \frac{\zeta(2)^3}{24\gamma^3},$$

and

$$\begin{aligned} D_0 &= \lambda^2 \left(a_2 - \frac{a^2}{2t} \right) + \lambda^3 \left(e + a_3 - \frac{aa_2}{t} + \frac{a^3}{3t^2} \right), \\ R_2 &= \alpha_2 + \alpha_1^2, \quad R_3 = \alpha_3 + 3\alpha_1\alpha_2 + \alpha_1^3, \\ R_4 &= \alpha_4 + 4\alpha_1\alpha_3 + 3\alpha_2^2 + 6\alpha_1^2\alpha_2 + \alpha_1^4. \end{aligned}$$

Then the exact formula generated by (5.22) is

$$\begin{aligned} C_2 &= D_0 + \frac{1}{2}\beta_0^2 + \beta_1 + \alpha_1\beta_0 + \frac{t}{2}(\beta_2 + 2\alpha_1\beta_1 + R_2\beta_0) \\ &\quad - \alpha_1 + \frac{2-t}{2}R_2 + \frac{5t}{6}R_3 + \frac{t^2}{8}R_4. \end{aligned} \quad (5.23)$$

The accompanying symbolic program expands (5.23) into a polynomial of degree at most two in t and four in λ and checks the first-correction cancellation by exact algebra.

5.6 Exact identities for diagonals of the residue array

The existing meromorphic moment transform has residues determined by

$$A_{k,m} = \frac{1}{k} [x^{k-1}] \Gamma(1+x)^k B(x)^m. \quad (5.24)$$

The joint transition has an exact algebraic counterpart in their near-diagonal coefficients.

Proposition 5.4 (Diagonal residue polynomials). *Put*

$$P_j(m) = (m+j+1)\gamma^{-m} A_{m+j+1,m}, \quad H(x) = \frac{\Gamma(1+x)B(x)}{\gamma x}.$$

Then P_j is a polynomial in m of degree j , with leading coefficient $c^j/j!$, and

$$P_j(m) = [x^j] \Gamma(1+x)^{j+1} H(x)^m. \quad (5.25)$$

Write $\ell_r = [x^r] \ell(x)$ and $h_r = [x^r] \log H(x)$. An explicit finite identity is

$$P_j(m) = \sum_{\nu_1+2\nu_2+\dots+j\nu_j=j} \prod_{r=1}^j \frac{(mh_r + (j+1)\ell_r)^{\nu_r}}{\nu_r!}. \quad (5.26)$$

In particular,

$$\begin{aligned} P_0(m) &= 1, & P_1(m) &= mc - 2\gamma, \\ P_2(m) &= \frac{1}{2}m^2c^2 + m\{\zeta(2)/2 + d - 3\gamma c\} + \frac{3}{2}\zeta(2) + \frac{9}{2}\gamma^2. \end{aligned} \quad (5.27)$$

If $g(w)$ is the inverse germ of $x/\Gamma(1+x)$, then

$$\sum_{j \geq 0} P_j(m) w^j = g'(w) \left(\frac{B(g(w))}{\gamma w} \right)^m. \quad (5.28)$$

These identities are identities of formal power series, with no assertion about convergence of the divergent full moment pole expansion.

Proof. Insert $k = m + j + 1$ in (5.24) and remove the factor $(\gamma x)^m$. This proves (5.25). Expansion of the exponential of $m \log H(x) + (j+1)\ell(x)$ proves (5.26). The coefficient of x in $\log H$ is $c > 0$, so the leading coefficient in m is exactly $c^j/j!$. The first two further coefficients yield (5.27).

Let $W_m(x) = \int_0^x B(u)^m du$. Lagrange–Bürmann inversion gives

$$W_m(g(w)) = \sum_{k \geq 1} A_{k,m} w^k.$$

Differentiate and divide by $\gamma^m w^m$. The result is precisely (5.28). $\square$

For fixed j , the contribution of the residue at $m + j + 1$, normalized by the leading scale in (5.2), is

$$P_j(m) \left(\frac{m+1}{m+j+1} \right)^{n+1} = \frac{(c\lambda)^j}{j!} + O_{S,j}(t/m).$$

The exponential profile is therefore the natural resummation of the near-diagonal residue directions. This observation explains the answer; the weighted integral proof above is what justifies the uniform limit. It would be invalid to obtain that limit by simply summing the divergent infinite fixed- m pole expansion.

5.7 Numerical diagnostics

The two numerical implementations in the package integrate respectively in $y = -\log x$ and directly in x , splitting at the moving saddle. They agree to the working precision of 60 decimal digits in the eight recorded cases. The calculations are high-precision diagnostics, not rational or interval enclosures. The analytic proofs establish the error orders. Table 1 records the central transition sequence; rounding of n is incorporated in the actual t and λ in each formula.

m	n	$\mathcal{R}_{n,m}$	Error after C_1	Error after C_2
30	105	2.427573439030	$1.0791 \cdot 10^{-2}$	$3.9071 \cdot 10^{-3}$
100	465	2.376317091692	$1.8754 \cdot 10^{-3}$	$2.0084 \cdot 10^{-4}$
300	1717	2.352781323812	$3.9099 \cdot 10^{-4}$	$1.4153 \cdot 10^{-5}$
1000	6915	2.341576352534	$6.1750 \cdot 10^{-5}$	$7.4413 \cdot 10^{-7}$

Table 1: Numerical diagnostics for n nearest to $(m+1)\log m$. Errors are the computed ratio minus the indicated approximation. No monotone improvement for every parameter and truncation is asserted.

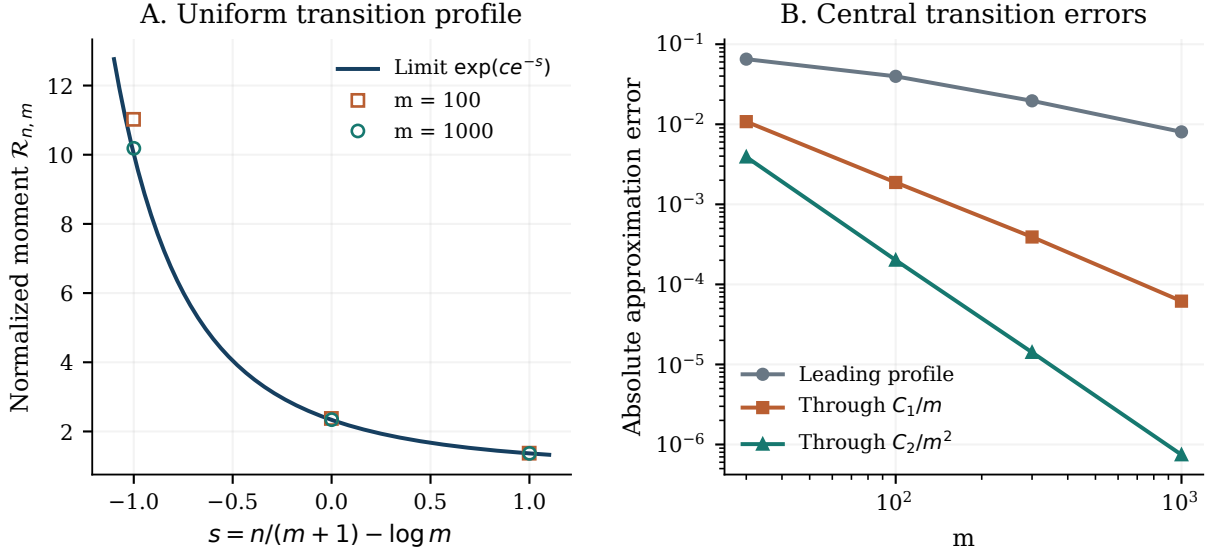


Figure 3: The proved profile $\exp(ce^{-s})$ and numerical checks near the central transition. The second panel shows errors of the successive approximations along the four cases in Table 1. The curves visualize the theorem and its diagnostics; they supply no independent proof.

6 A sharp formal obstruction at weight six

The audited manuscript proves a complementary-depth bound for one-variable multiple polylogarithms: after adjoining a transformed argument and endpoint constants, a weight- w , depth- d value has a representation of reduced depth at most $w - d$. The first critical triple weight is therefore six. The following result identifies exactly what survives if one tries to reduce every such triple further using shuffle products and the six standard argument transformations. It is a theorem about a specified formal quotient, with no assertion of numerical period independence.

6.1 The quotient and the argument action

Let $V = \mathbb{Q}x + \mathbb{Q}y$, let $\text{Sh}(V)$ be its shuffle algebra, and let J be the augmentation ideal. Set

$$Q_6 = (J/J^2)_6.$$

The depth of a binary word is its number of y letters. Including words with trailing x makes linear substitutions immediate; shuffle normalization with $H_x(z) = \log z$ removes such trailing letters without changing the positive-weight indecomposable quotient. The usual outer-letter-first interpretation is

$$x = \frac{dt}{t}, \quad y = \frac{dt}{1-t}, \quad H_{0^{s_1-1}1^{s_2-1}\dots 0^{s_d-1}1}(z) = \text{Li}_{s_1, \dots, s_d}(z, 1, \dots, 1),$$

where 0 denotes x and 1 denotes y .

The group of six Möbius transformations permuting $0, 1, \infty$ acts on the letter space through the matrices

$$R = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}. \quad (6.1)$$

Matrices act on columns, so $R(x) = -y$, $R(y) = -x$, whereas $M(x) = x + y$, $M(y) = -y$. These are the pullbacks for $z \mapsto 1 - z$ and $z \mapsto z/(z - 1)$, respectively. The six matrices are I, R, M, RM, MR, RMR . Landen substitution fixes the initial endpoint zero. Complementation changes it to one; ordinary path composition supplies endpoint constants and products of lower-weight integrals. Thus the same letter action models these argument changes modulo products and endpoint constants. The formal space is defined independently of any special-point evaluation.

Theorem 6.1 (Exactly one missing triple direction). *Let $D_2 \subset Q_6$ be the span of words of depth at most two, and let E_2 be the span of their images under the six matrices above. Then*

$$\dim Q_6 = 9, \quad \dim E_2 = 8.$$

The span is unchanged if all of $\text{GL}_2(\mathbb{Q})$ is allowed in place of the six matrices. Its annihilator is generated by coefficient pairing with the Lie polynomial

$$\Omega = [[x, [x, y]], [y, [x, y]]]. \quad (6.2)$$

On the three triple generators used in the manuscript this pairing is

<i>word</i>	000111	001011	001101
<i>index</i>	(4, 1, 1)	(3, 2, 1)	(3, 1, 2)
$\langle \Omega, \text{word} \rangle$	0	-1	2.

(6.3)

In particular, the class of (3, 2, 1) supplies the single additional formal direction.

Proof. The coefficient pairing is $\langle \sum a_w w, \sum b_w w \rangle = \sum a_w b_w$. Lie polynomials are primitive for the coproduct dual to shuffle multiplication: this follows from $\Delta x = x \otimes 1 + 1 \otimes x$, the same formula for y , and closure of primitive elements under commutators. Hence every Lie polynomial pairs to zero with every shuffle product of two nonempty words. Consequently Ω defines a functional on Q_6 .

Every word in Ω has exactly three x and three y letters, so the functional vanishes on D_2 . It is nonzero: the coefficient of 001011 is -1 , and expansion gives the three entries in (6.3). There is also a covariance under every invertible linear substitution N . Put $C = [x, y]$, $A = [x, C]$, and

$B = [y, C]$. The element C transforms by $\det N$, while the pair (A, B) transforms by $\det N$ times the original two-dimensional matrix. Their commutator therefore satisfies

$$N(\Omega) = (\det N)^3 \Omega. \quad (6.4)$$

Since N and its transpose have the same determinant,

$$\langle \Omega, N(w) \rangle = \langle N^T(\Omega), w \rangle = (\det N)^3 \langle \Omega, w \rangle.$$

Thus Ω annihilates the full linear-letter orbit of D_2 . Both proposed spans have codimension at least one.

For the reverse inequality, the Lyndon description of the shuffle algebra [7] gives the binary Witt count

$$\dim Q_6 = \frac{2^6 - 2^3 - 2^2 + 2}{6} = 9.$$

The nine Lyndon words are

$$\begin{aligned} &000001, 000011, 000101, 000111, 001011, 001101, \\ &001111, 010111, 011111. \end{aligned}$$

For each Lyndon word l , let $b(l)$ denote its standard Lie bracketing: if $l = uv$ and v is its longest proper Lyndon suffix, put $b(l) = [b(u), b(v)]$, with the letters as initial cases. These nine primitive elements form a dual family of functionals on Q_6 .

Take, in order, the eight word classes

$$\begin{aligned} &I(000001), I(000011), I(000101), \\ &R(000001), R(000011), R(000101), \\ &M(000001), M(000101), \end{aligned}$$

and pair them with the eight $b(l)$ obtained by omitting 001101 from the ordered Lyndon list. Direct commutator expansion gives

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4 & -1 & 0 \\ -1 & 1 & 0 & -1 & 0 & 1 & 0 & -1 \\ 0 & -2 & 1 & 4 & 1 & -7 & -1 & 11 \end{pmatrix}. \quad (6.5)$$

Its determinant is 1, proving independence of the eight classes. The six-transformation span is consequently the eight-dimensional kernel of Ω . The full linear-letter span lies between that space and the same kernel, proving the stronger assertion too. $\square$

6.2 Two exact quotient identities

The rank statement also supplies concise constructive reductions. Write $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$ and $F_{a,b,c}(z) = \text{Li}_{a,b,c}(z, 1, 1)$, and put $u = 1 - z$, $v = z/(z - 1)$. Here $\equiv$ denotes equality modulo products of positive-weight hyperlogarithms and endpoint constants, of total weight six. The identities use the common analytic continuation from $0 < z < 1$.

Proposition 6.2. *One has*

$$F_{4,1,1}(z) \equiv -\text{Li}_6(z) + F_{5,1}(z) + \text{Li}_6(u) - F_{5,1}(u) - \text{Li}_6(v), \quad (6.6)$$

$$\begin{aligned} 2F_{3,2,1}(z) + F_{3,1,2}(z) &\equiv -\text{Li}_6(z) + F_{5,1}(z) + F_{4,2}(z) \\ &\quad - 10\text{Li}_6(u) + 10F_{5,1}(u) + F_{4,2}(u) \\ &\quad - \text{Li}_6(v) - F_{4,2}(v). \end{aligned} \quad (6.7)$$

Proof. Expand the nine rows obtained by applying I, R, M , in that order, to 000001, 000011, 000101. The coefficient vectors for (6.6) and (6.7), respectively, are

$$(-1, 1, 0, 1, -1, 0, -1, 0, 0), \quad (-1, 1, 1, -10, 10, 1, -1, 0, -1).$$

Subtract each corresponding word combination from its left side. Pairing the residual with each of the nine standard Lyndon Lie polynomials gives zero, by direct expansion. The primitive space is dual to the indecomposable quotient, so each residual belongs to J^2 . Landen pullback and the endpoint-transport formula then give the stated analytic congruences. This is an equality in the specified quotient; no product term is silently asserted to vanish as a function. $\square$

The supplied standard-library verifier reconstructs all of these objects, checks the determinant-one minor, verifies the two congruences, and checks Ω against all 320 nonempty binary shuffle products of weight six. Every operation uses exact integers or rational numbers. No numerical polylogarithm value or rank estimate enters.

Consequences and limits. The general complementary-depth bound three is sharp for this formal class at weight six, even after the argument alphabet is enlarged by all six standard transformations. Exactly one triple coordinate remains, rather than an unspecified failure of a search. At a particular cyclotomic point, additional identities can still kill that coordinate. The theorem therefore proves neither numerical irreducibility of $F_{3,2,1}(i)$ nor an obstruction to identities using other color alphabets. The existing octahedral S_4 proof illustrates why enlarging the available laws can overcome a restricted formal obstruction. The manuscript’s S_6 candidate remains conjectural.

7 Corrections and status revisions for the manuscript

The source’s discovery chapter already gives a careful distinction between experimental evidence, exact relation matrices, and numerical period identities. A few earlier passages should be brought into line with that standard. The proposed patch is restricted to the pinned source and is supplied for review; it has not been applied to the remote repository.

7.1 A definite bibliographic correction

The plastic-base discussion in `chapters/03-algebraic.tex` describes a sequel of Abouzahra–Lewin as never published. The sequel was published: M. Abouzahra, L. Lewin, and Xiao Hongnian, *Polylogarithms in the field of omega (a root of a given cubic): Functional equations and ladders*, *Aequationes Mathematicae* **33** (1987), 23–45 [4]. The publisher records the issue date as February 1987 and the DOI as 10.1007/BF01836149. Its abstract distinguishes analytically derived lower-order results from numerically found higher-order ladders. Restoring this reference therefore improves both attribution and proof-status tracking.

The later literature also includes Gangl’s functional equations and proved ladders in weights six and seven [9]. Those statements use a single-valued modified polylogarithm; the projection is

the real part in odd weight and the imaginary part in even weight. In particular, an even-weight modified identity at real arguments cannot simply be read as an identity for ordinary real Li_{2k} . The correction patch makes this normalization distinction explicit.

7.2 Numerical confirmation is a distinct status

The golden-ladder discussion uses headings and phrases such as *conjectures*, *verified*, *also hold*, and *are correct* beside finite-precision residual checks. A small computed residual establishes numerical consistency with a formula. By itself it does not establish an identity of periods. The proposed edits say *numerically confirmed* where that is the evidence actually provided, and retain analytic or exact status when a proof is present.

This is a local editorial correction, not a claim that every displayed historical ladder lacks a proof in the literature. A formula with a published proof should be matched to that proof, including its normalization and branches. A proposed specialization of a functional equation should include the exact substitution and residual cancellation. In particular, citing a family of Kummer equations without supplying or identifying the required specialization is less specific than a replayable derivation.

The new seed theorem sharpens the source's support question but does not remove this distinction. It proves that no new positive even golden power adds a multiplicative seed. It does not prove the constants in each higher ladder or a maximal ladder weight of nine. The remaining question concerns compatibility with functional equations, products, and regulator data on an already complete seed space.

7.3 Restoring a missing canonical ladder definition

The canonical elimination chain in the same source uses L_{12} without defining it. Its logarithmic tail also contains factorials with negative indices at the low orders where vanishing is discussed. The proposed repair omits those summands and reconstructs the missing notation from the first displayed weight-five formula. Put $\ell = \log \rho$ and, for $n \geq 1$, define

$$\text{tw}_{12}(n) = -\frac{13}{48} \frac{\ell^n}{n!} + \frac{1}{48} \zeta(2) \frac{\ell^{n-2}}{(n-2)!} - \frac{19}{1728} \zeta(4) \frac{\ell^{n-4}}{(n-4)!}, \quad (7.1)$$

$$L_{12}(n) = \frac{\text{Li}_n(\rho^{12})}{12^{n-1}} - \frac{3}{2} \frac{\text{Li}_n(\rho^6)}{6^{n-1}} - \frac{\text{Li}_n(\rho^4)}{4^{n-1}} + \frac{11}{48} \frac{\text{Li}_n(\rho^2)}{2^{n-1}} + \text{tw}_{12}(n). \quad (7.2)$$

The negative-factorial convention applies in (7.1). If E_1, E_2, E_3 are the left-minus-right residuals of the three displayed source weight-five formulas, the exact rational normalizations are

$$\begin{aligned} L_{12}(5) - \frac{67}{6912} \zeta(5) &= \frac{E_1}{15 \cdot 12^4}, \\ L_{20}(5) - \frac{201}{10000} \zeta(5) &= \frac{E_2}{9 \cdot 20^4}, \\ L_{24}(5) + \frac{1541}{110592} \zeta(5) &= \frac{E_3 + (64/3)E_1}{-15 \cdot 24^4}. \end{aligned} \quad (7.3)$$

Thus the canonical L_{24} requires a recombination before rescaling. These equalities follow by exact rational coefficient comparison, and restore a reviewable meaning to the source's elimination chain. They do not assert $E_j = 0$ without a proof of the underlying evaluations.

The replay program checks all three transformations exactly and compares Li_n from `mpmath` with a separate 600-term defining series at 160 working digits. The largest recorded canonical residual

over orders one through five is below 1.12×10^{-161} . The series has an analytic truncation bound; floating-point rounding is not interval-enclosed. The notation repair and the numerical evaluation therefore retain their different logical statuses.

7.4 Limits that should remain explicit

The fixed- m moment theorem in the source is sound in its stated regime. The factor $e^{ce^{-s}}$ in Theorem 5.1 explains why extending that regime without a uniform analysis would be incorrect. The source’s broader question about simultaneous large exponents is only partially resolved: the present strip is near $n \sim m \log m$, and does not cover every fixed ratio n/m .

For real-order depth-two functions, the equality $a + b = 1$ changes the meaning of a unit-circle value. The coefficients then tend to $1/a$, so individual terms of the boundary series do not tend to zero. The signed-measure continuation gives well-defined analytic values away from the cut; ordinary series convergence must not be asserted.

Finally, the S_4 identity is already proved in the pinned source. Its S_6 identity has total weight seven and remains conjectural. The source also already proves that two displayed S_6 coordinate forms are equivalent. That exact coordinate change is not a proof that either conjectural residual vanishes.

8 Further research questions and proposed relations

The following program separates promising conjectures from questions for which even the correct global statement remains uncertain. The priority order reflects how directly each task builds on a proved result in this article.

8.1 From complete seeds to analytic golden ladders

The complete even-power alphabet is now

$$\{\rho^2, \rho^4, \rho^6, \rho^8, \rho^{10}, \rho^{12}, \rho^{20}, \rho^{24}\}.$$

The next task is to obtain explicit analytic certificates for the source’s higher golden formulas on this finite set. A useful certificate would list the functional equations used, the algebraic substitutions, every lower-weight product contribution, and the constant fixed by evaluation at a regular point. The six integral seed vectors provide canonical input rows, and the exterior-symbol calculation supplies an exact first obstruction at every weight.

Three questions are especially concrete. Which of the four even symbol directions extend through each weight? Can the regulator or coaction constraints be expressed in a finite sequence of exact matrices? If extension ceases in a specified ladder class, can one prove that failure inside that class? A numerical absence of another ladder, or the completeness of its argument list, is not such a proof.

8.2 Classifying all quadratic bases with nontrivial seeds

The uniform cutoff reduces any positive real quadratic unit to at most twenty-four indices. One can therefore seek a classification in terms of its trace, norm, and power within the unit group. For a fixed base, ideal factorization followed by an integral nullspace and unit calculation gives a finite algorithm. The stronger family problem is to determine all bases for which the nullspace is nonzero, without enumerating an unbounded range of traces.

The golden ranks $6, 4, 1, 1, 0, \dots$ show why retaining the base, not merely its number field, matters. Extending this classification to cubic or quartic units will require a different effective primitive-divisor analysis and careful treatment of the larger unit rank. The plastic examples are a particularly relevant first case because their published functional equations are already available [4]. Campbell's more recent construction of dilogarithm families with unboundedly many terms in the defining base polynomials [10] supplies a complementary direction in which the degree of the base is allowed to grow. A useful comparison would separate that variable-degree construction from the fixed-degree primitive-divisor bounds used here.

8.3 A proposed strict angular monotonicity

Theorem 3.3 proves $x'(s) > 0$, but the angle also depends on the changing radius. The stronger statement is the following plausible continuation.

Conjecture 8.1 (Radial angular monotonicity). *For every $a, b > 0$ with $a + b \geq 1$,*

$$\frac{d}{dR}\theta_{a,b}(R) < 0 \quad (0 < R \leq 1).$$

Equivalently, the analytic abscissa from Theorem 3.3 satisfies

$$2s x'_{a,b}(s) > x_{a,b}(s) \quad (0 < s \leq 1).$$

The small-radius expansion proves the assertion for sufficiently small R , for each fixed pair a, b . It holds exactly when $a = b = 1$, and the recorded numerical samples show decreasing angles at the four tested radii for each of four parameter pairs. That is limited evidence, not a proof over the parameter domain. The existing sign argument establishes positivity of G_x and negativity of G_s ; it does not establish the stronger comparison $-2sG_s > xG_x$ required here. A proof may need a further inequality for the cumulative signed measure, or a counterexample may reveal the correct restricted domain.

Separate questions concern monotonicity in a and b . The strictly decreasing density-crossing location in Corollary 3.7 does not settle them. A likelihood-ratio comparison for the cumulative measures would give more useful information than comparing the densities' zeros alone.

8.4 Below the finite-measure threshold and at its corners

For $a + b < 1$, the coefficients grow like $n^{1-a-b}/(1-b)$. Finite signed measures are therefore impossible, but the generating function and its analytic continuation remain meaningful. Can a controlled derivative or fractional derivative of a measure replace the representation used here? Does the unique angular zero survive throughout that larger domain, and what happens to it as $a + b \downarrow 0$? These are open questions in this continuation, not consequences of the signed-measure theorem.

At the critical boundary, Section 4 fixes b in $(0, 1)$. Joint limits $b \downarrow 0$ or $b \uparrow 1$ can change the scale and the constants. Determining a uniform description in those corners would connect power-law density crossings with logarithmic regimes. The proved exponential concentration law also suggests using its rescaled coordinate for quadrature, with explicit error bounds that remain stable as $a + b \downarrow 1$.

8.5 The full two-exponent moment problem

The integral obeys the exact symmetry $M_{n,m} = M_{m,n}$. The present transition describes one highly asymmetric region, and reflection gives the opposite region. A complete asymptotic theory should

connect these regions with the interior saddle for bounded positive n/m . A starting point is the phase

$$\Phi_\alpha(x) = \alpha \log f(x) + (1 - \alpha) \log f(1 - x), \quad \alpha = \frac{n}{n + m}.$$

Before asserting a uniform saddle formula, one should prove the number and nondegeneracy of its maxima as α varies. The central symmetric case and possible transitions in the saddle structure merit separate analysis.

Another question is how far the proved critical strip can widen with m . The weighted localization proof identifies two competing requirements: the endpoint amplitude must remain uniformly expandable, and the tilted gamma tails must remain negligible. An explicit maximal window, or several matched windows, would be more informative than an unspecified two-variable asymptotic.

8.6 A structural conjecture for all correction coefficients

The construction in Theorem 5.3 initially produces rational functions of t . Exact simplification removes every negative power of t from C_1 and C_2 . The following is an exploratory coefficient identity, with evidence currently limited to those two nontrivial orders.

Conjecture 8.2 (Polynomial correction structure). *For every $j \geq 1$, the coefficient defined by (5.22) belongs to*

$$\mathbb{Q}[\gamma, \gamma^{-1}, \zeta(2), \dots, \zeta(j + 2)][t, \lambda], \quad \deg_t C_j \leq j, \quad \deg_\lambda C_j \leq 2j, \quad C_j(t, 0) = 0.$$

The theorem already proves the growth bound $C_j = O(t^j)$ for bounded λ ; the additional assertion is exact cancellation of inverse powers and a uniform polynomial degree in λ . The diagonal residue identity (5.28) suggests a combinatorial mechanism. Turning that suggestion into a coefficientwise argument would avoid an invalid summation of the full divergent pole expansion. A useful intermediate deliverable is an exact C_3 or C_4 certificate obtained both from the gamma cumulants and from finite residue diagonals.

8.7 Lifting the depth congruences and resolving S_6

The two congruences in Proposition 6.2 can be lifted to complete functional identities by decomposing their word residuals into shuffle products and transporting the endpoints. An explicit lifted formula should contain every product and endpoint constant, with branches fixed, and should be checked by differentiation and one base value. This would make the formal reduction directly usable for Gaussian special values.

At higher weights, compute the annihilators of the low-depth GL_2 -orbit in the free Lie coalgebra. The determinant-cube covariant at weight six is a particularly small example. A representation-theoretic classification would replace unstructured searches by a prediction of the missing directions, while keeping evaluations at cyclotomic points separate.

For clarity, a concrete pre-existing target is retained below. Let $G = \beta(2)$, let $\beta(s) = \sum_{n \geq 0} (-1)^n / (2n + 1)^s$, and put

$$S_6 = \sum_{n \geq 0} \frac{(-1)^n H_n}{(2n + 1)^6}, \quad g_{a,b} = \Im \mathrm{Li}_{a,b}(i, 1).$$

The pinned manuscript conjectures the weight-seven identity

$$\begin{aligned} S_6 = & \frac{722}{527} g_{6,1} + \frac{40}{527} g_{4,3} - \frac{128}{155} g_{2,5} + \frac{15191}{28569600} \pi^7 \\ & - \frac{3}{124} G \zeta(5) - \frac{2373}{10540} \beta(4) \zeta(3) - 2\beta(6) \log 2. \end{aligned} \tag{8.1}$$

This article does not prove (8.1). The exact equivalence with the source’s alternative coordinate form is already established there. A substantive next step is to find a convergent higher-weight analogue of the octahedral or Cayley relation used for S_4 , or to reduce the residual in a larger rigorously specified iterated-integral alphabet. Repeating integer-relation searches in equivalent coordinate systems supplies no missing analytic law.

9 Reproducibility and integration

The archive contains the full LaTeX source, the compiled article, standalone figures, executable verification programs, their recorded outputs, source provenance, and a proposed manuscript correction patch. The four main mathematical sections can be integrated independently; their labels have separate prefixes. The boundary section depends on the signed-kernel section.

The golden certificate uses only integer and rational arithmetic, checking six generating relations, two restricted-base identities, eighteen primitive prime-ideal witnesses, and the finite restriction ranks. Its independent audit checks every smaller residue exponent, not just the proper-power tests in the original script. The infinite support theorem additionally uses the cited primitive-divisor theorem.

The depth certificate reconstructs the Lie polynomial and all six letter substitutions. It verifies the determinant-one minor, the two quotient identities, and all 320 nonempty binary shuffle products at weight six. These are exact finite calculations.

The moment program derives C_1, C_2 by symbolic algebra and checks the gamma moments about the mode. Its numerical part compares direct x -quadrature with the transformed $y = -\log x$ integral in eight cases at 60 decimal digits of working precision. The geometric diagnostics compare two kernel integrals, test the critical density and deficit, and locate interior-disk zeros, including the exact circular example. The boundary diagnostics are documented alongside the additional local expansion. All floating-point outputs are labeled as numerical checks, not interval certificates or equality proofs.

The package’s `README.md` gives the build and replay commands; `integration/README.md` maps the sections to the pinned manuscript. The exact scripts require only the Python standard library; symbolic and numerical checks declare their additional dependencies. Source hashes and the pinned commit make the editorial patch reviewable against an unambiguous baseline.

The intended integration preserves the distinction between proved theorems, formal quotient statements, established external inputs, and the explicit conjectures in Section 8. The latter should remain in the research agenda until an appropriate proof or counterexample is supplied.

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